



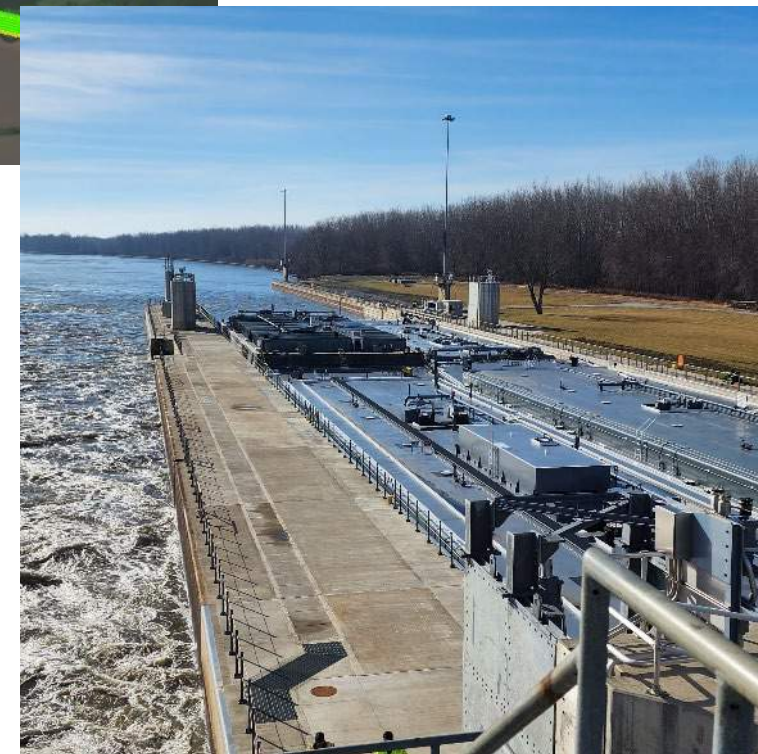
New 1200 Foot Lock at LaGrange Lock and Dam Industry Day

8 April 2025

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US Army Corps of Engineers,
Rock Island District
Clock Tower Building
PO Box 2004, Rock Island, IL



PROJECT PURPOSE

- **PURPOSE:**

- To improve efficiency, operability, reliability, and safety for navigation traffic transiting the facility.

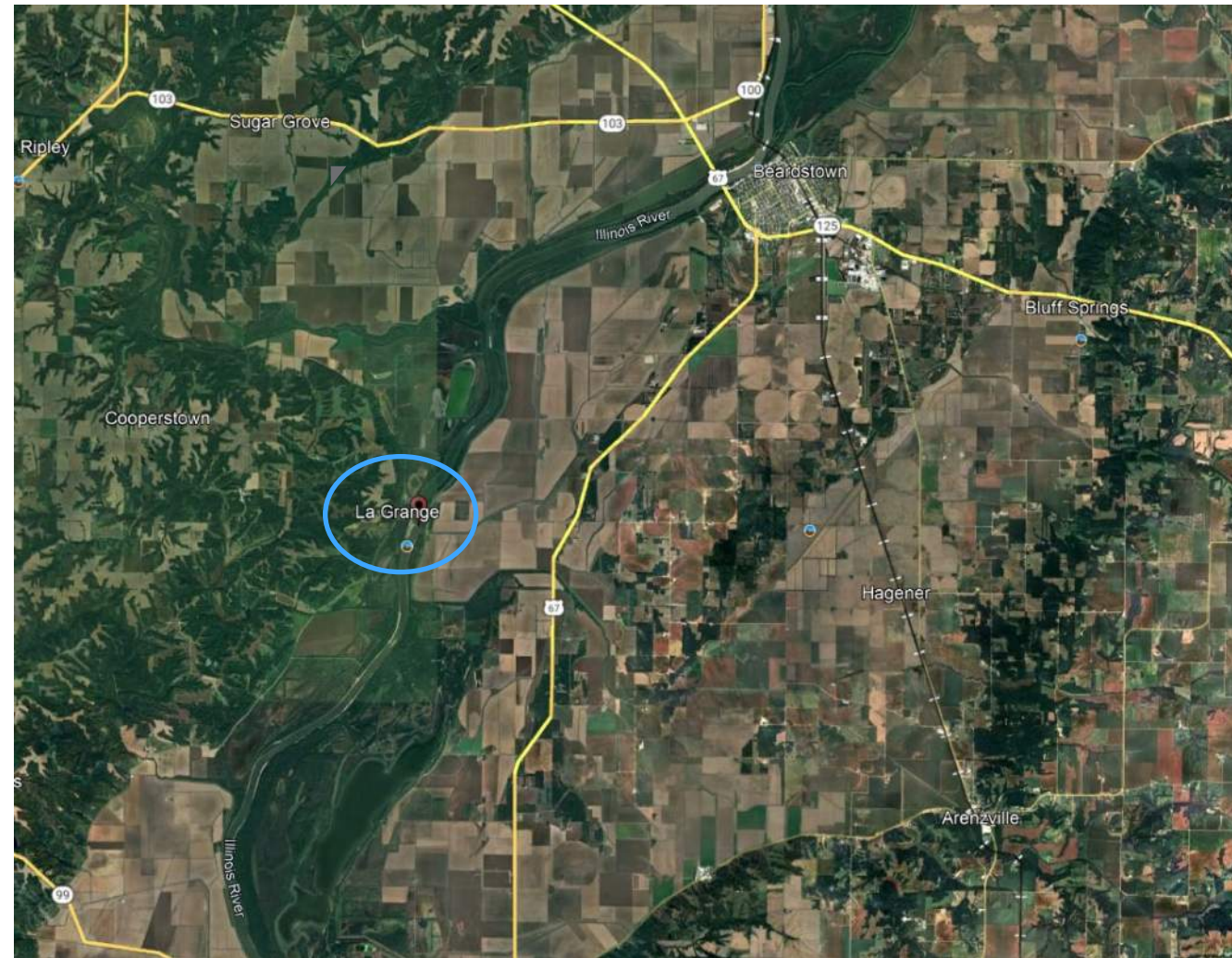
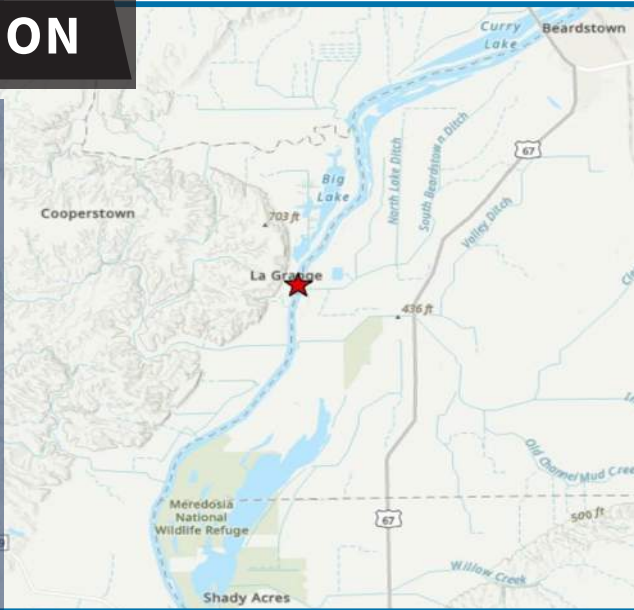
- **KEY FEATURES:**

- New 1,200-foot navigation lock chamber landward of existing 600-foot lock chamber
- New upstream and downstream approach channels
- New lock will be constructed with minimal disruption to navigation traffic
- Existing 600-foot lock remains in operation during construction of new lock

- **PROJECT BENEFITS:**

- **REDUCED LOCKAGE TIMES:** Lockages reduced from 2.5 hours to approximately 45 minutes, saving time and increasing efficiency.
- **INCREASED CAPACITY:** The new lock could accommodate the largest tow configurations on the Illinois Waterway, meeting the demands of the waterway.
- **ENHANCED RELIABILITY:** The existing 600-foot lock chamber would remain operational after the new lock is constructed, providing added redundancy and minimizing downtime.
- **IMPROVED SAFETY:** Professional mariners would no longer have to “split” tows to transit the chamber, reducing the risk of accidents and enhancing overall safety.

PROJECT LOCATION



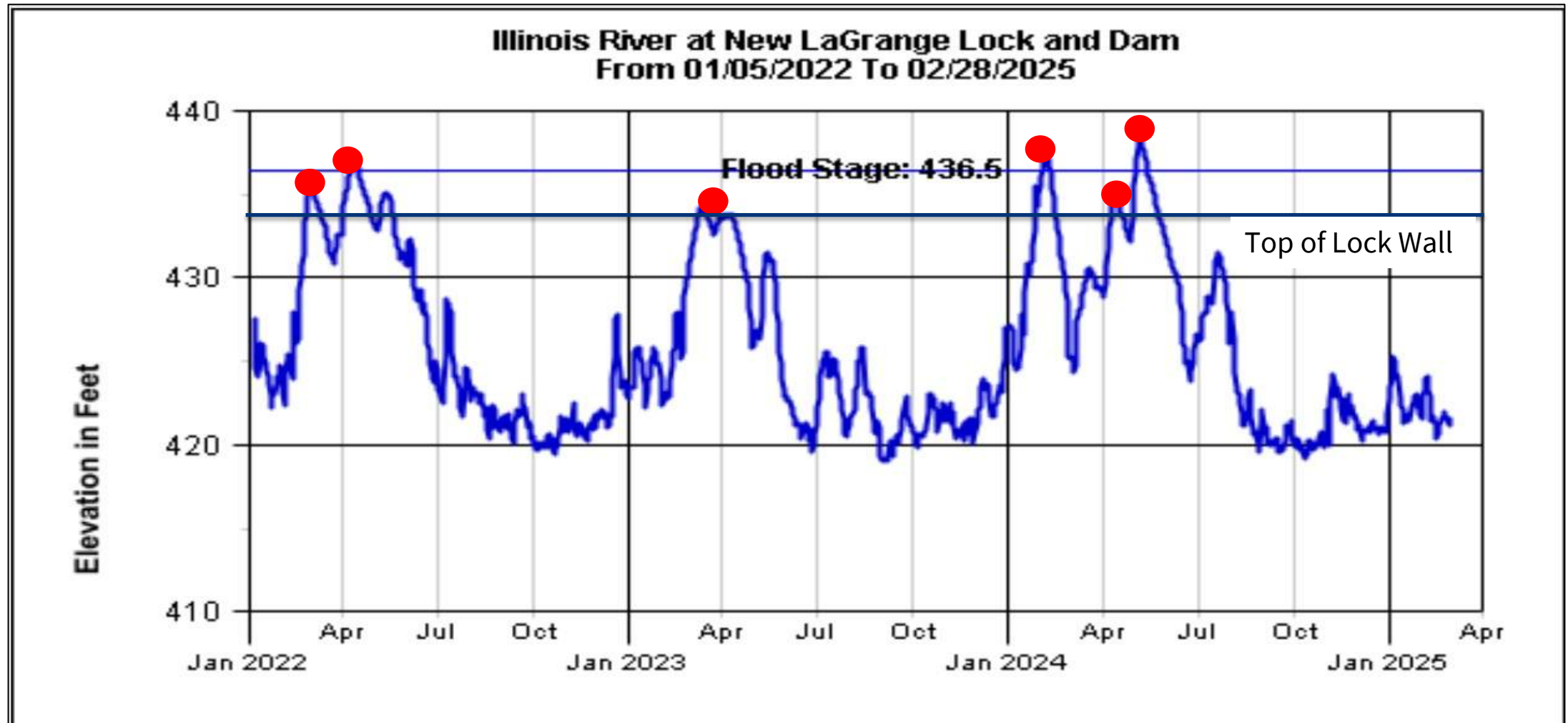
EXISTING 600FT LOCK AND DAM

- New 1,200-foot lock will be constructed in a cofferdam landward of the existing lock.
- Existing lock to remain operation during construction
- High water events at existing lock bring significant debris.



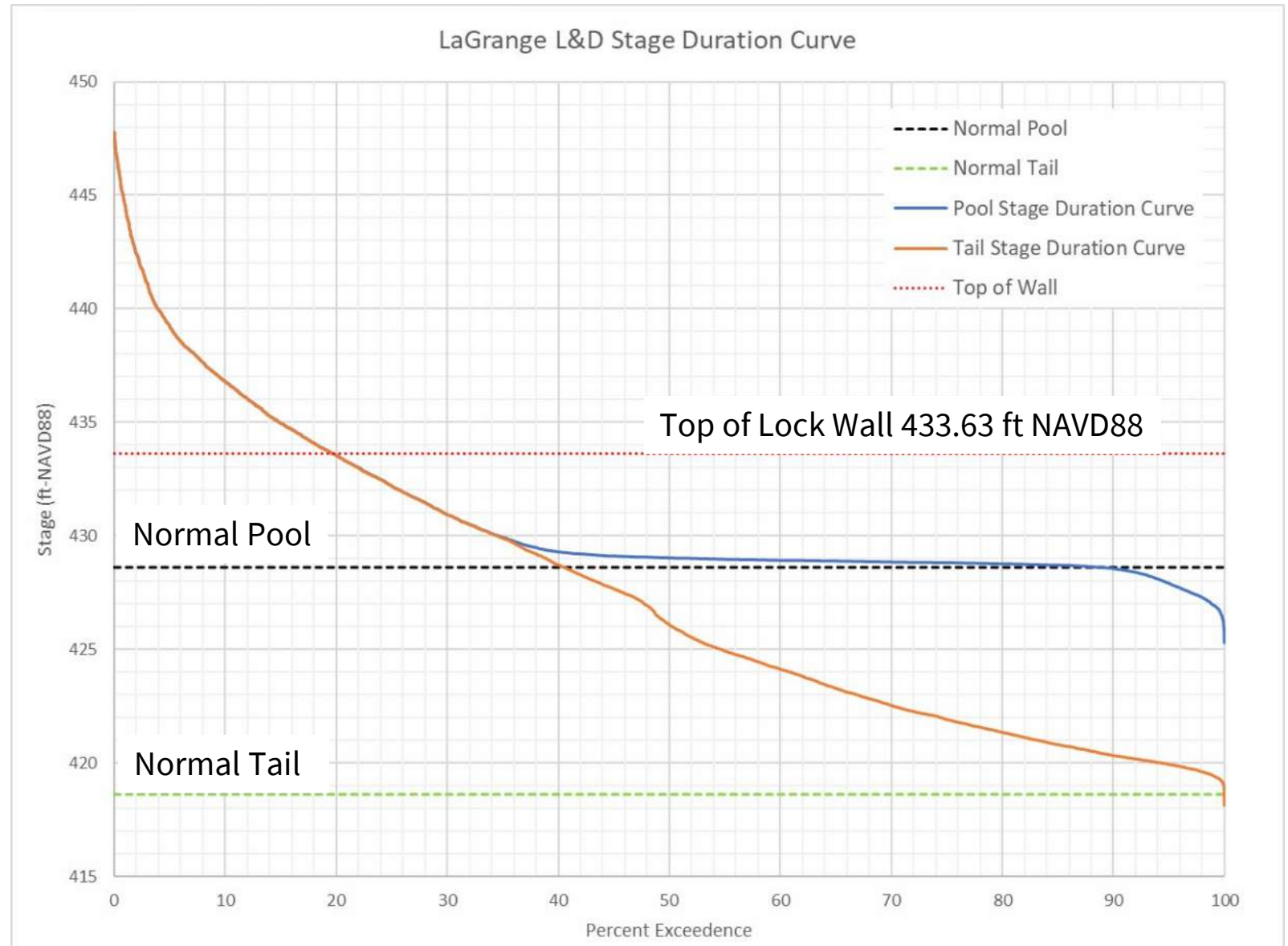
EXISTING LOCK WALL OVERTOPPING

- 6 Over Overtoppings in the last 3 years
- Top of Lock Wall – 433.63 ft NAVD88



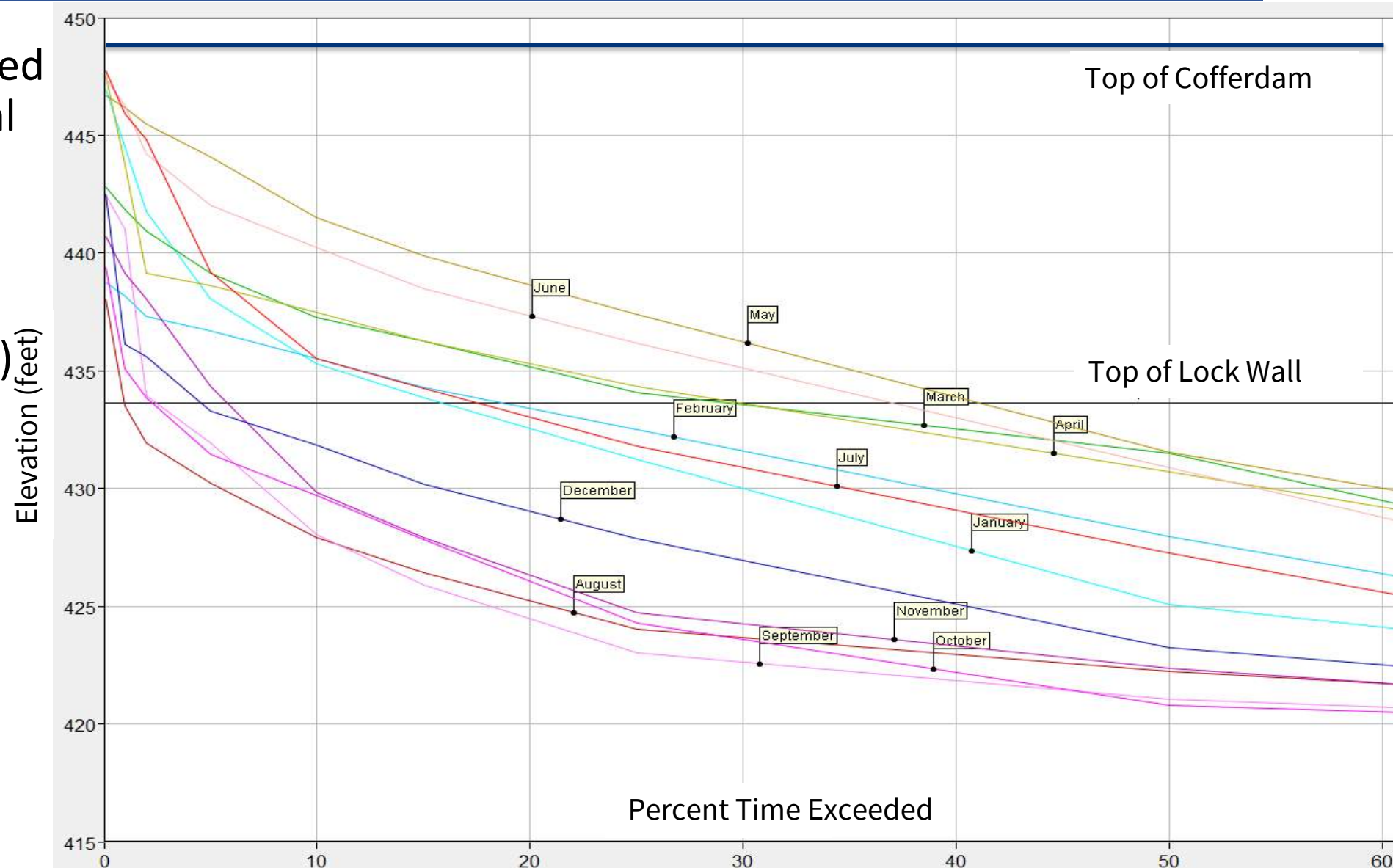
LAGRANGE L&D STAGE DURATION CURVE

Water is above the top of the lock wall approximately 20% of the time



LAGRANGE DURATION ANALYSIS PLOT

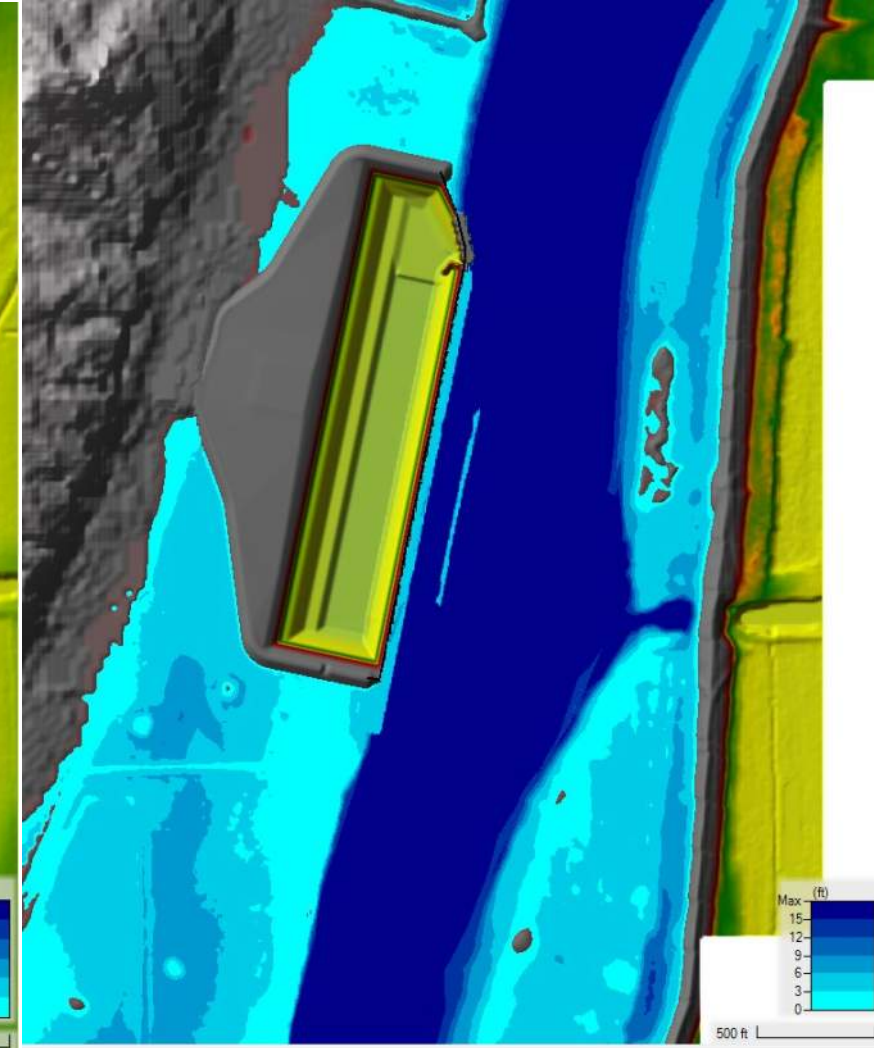
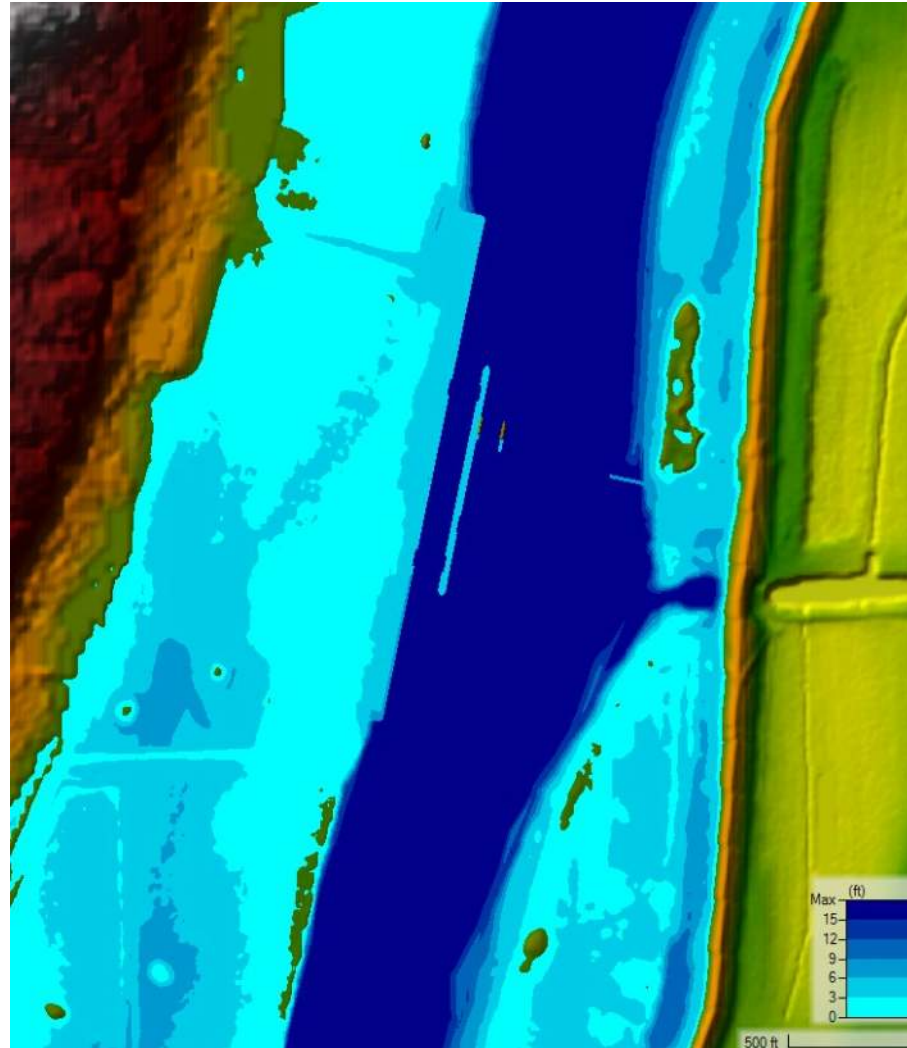
- Cofferdam height based on updated 1% annual chance exceedance
- Cofferdam Height – 449.1 ft NAVD88
- Flood of record (2015) – 447.8 ft NAVD88
- Top of lock wall – 433.63 ft NAVD88



Existing Conditions

Conditions with Cofferdam

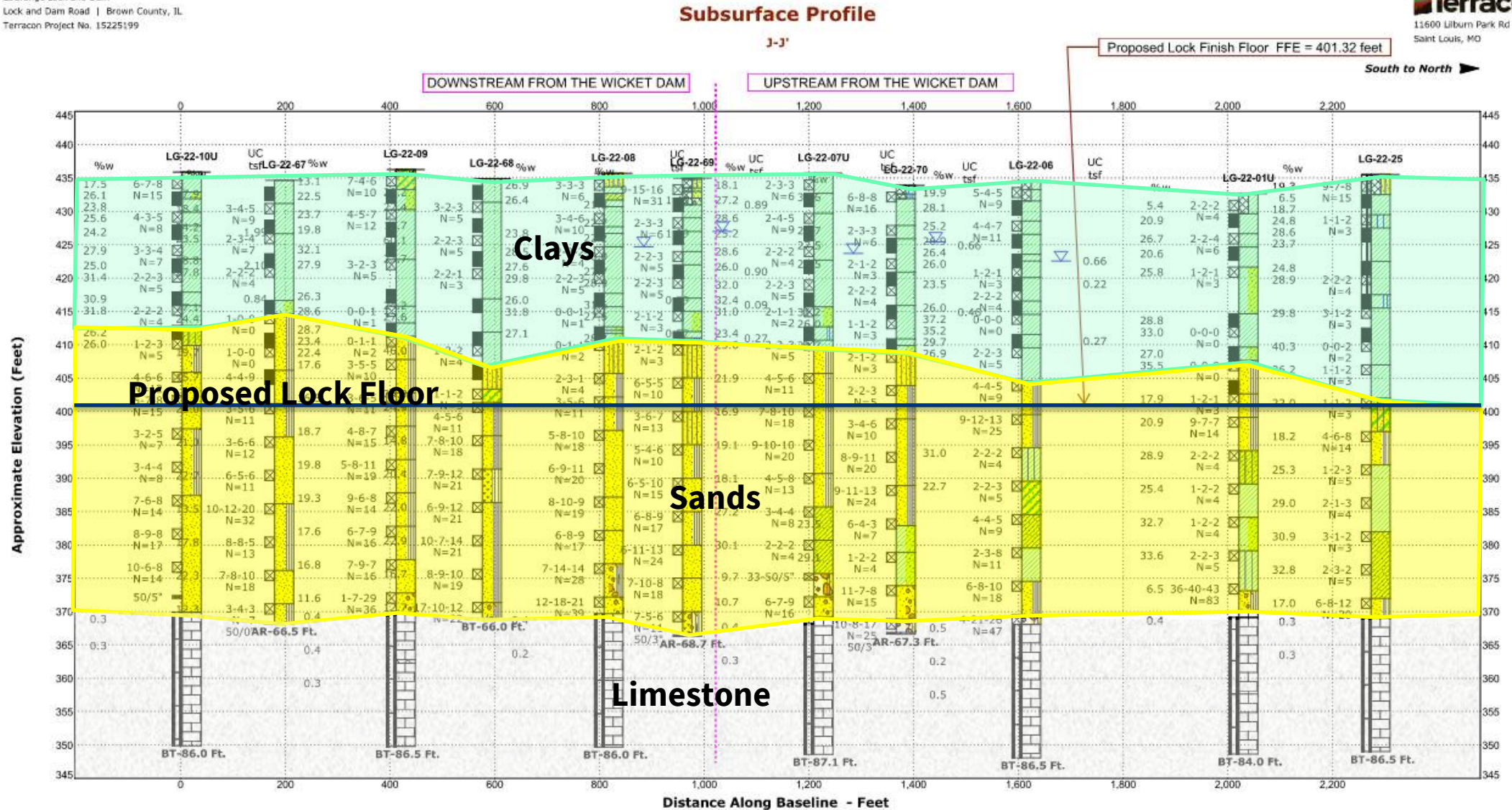
- Cofferdam height based on updated 1% annual chance exceedance
- Cofferdam Height – 449.1 ft NAVD88
- Flood of record (2015) – 447.8 ft NAVD88
- Inundation for 80,000 cfs
Approx. 50% Annual
Chance Exceedance



SITE GEOLOGY-NEW LOCK CENTERLINE

LaGrange Lock and Dam
Lock and Dam Road | Brown County, IL
Terracon Project No. 15225199

Terracon
11600 Lilburn Park Rd
Saint Louis, MO



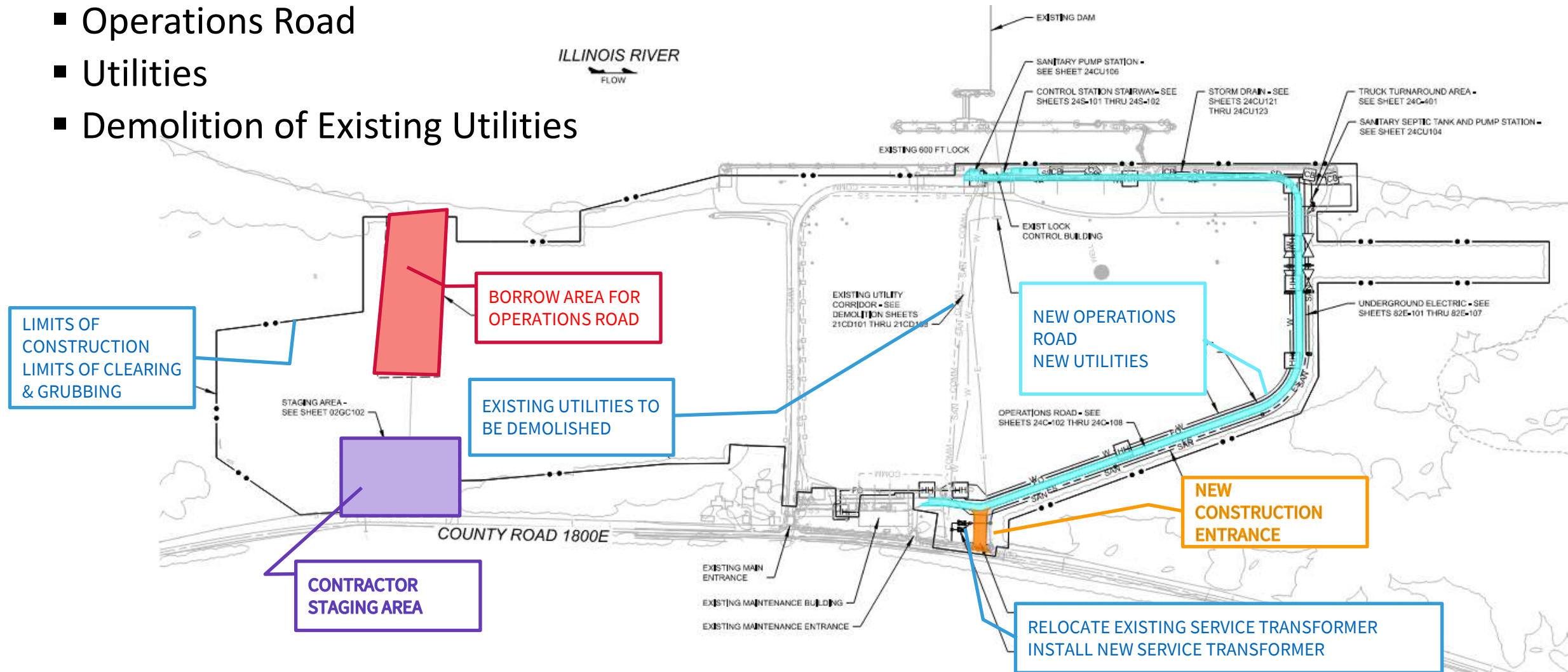
- CURRENTLY ON PATH TO COMPLETION IN 2032
 - Depends on funding stream remaining consistent
 - Received \$20 million for design in FY22
 - Received \$49.3 million for design & construction in FY23
- AE TASK ORDER
 - Site investigations – Complete
 - Survey, Geotech Explorations, Environmental Compliance Surveys – Complete
 - Charrettes – Complete
 - Pre-35% Design – Complete
- AE TASK ORDER TO COMPLETE DESIGN AWARD 9/1/23
- HYDRAULIC DESIGN 11/22 – 9/23
 - Configuration of lock, guide walls and approaches
 - Ice Mitigation
- AE Final Design – In Progress, 100% Final Design in February 2026

PROJECT FEATURES

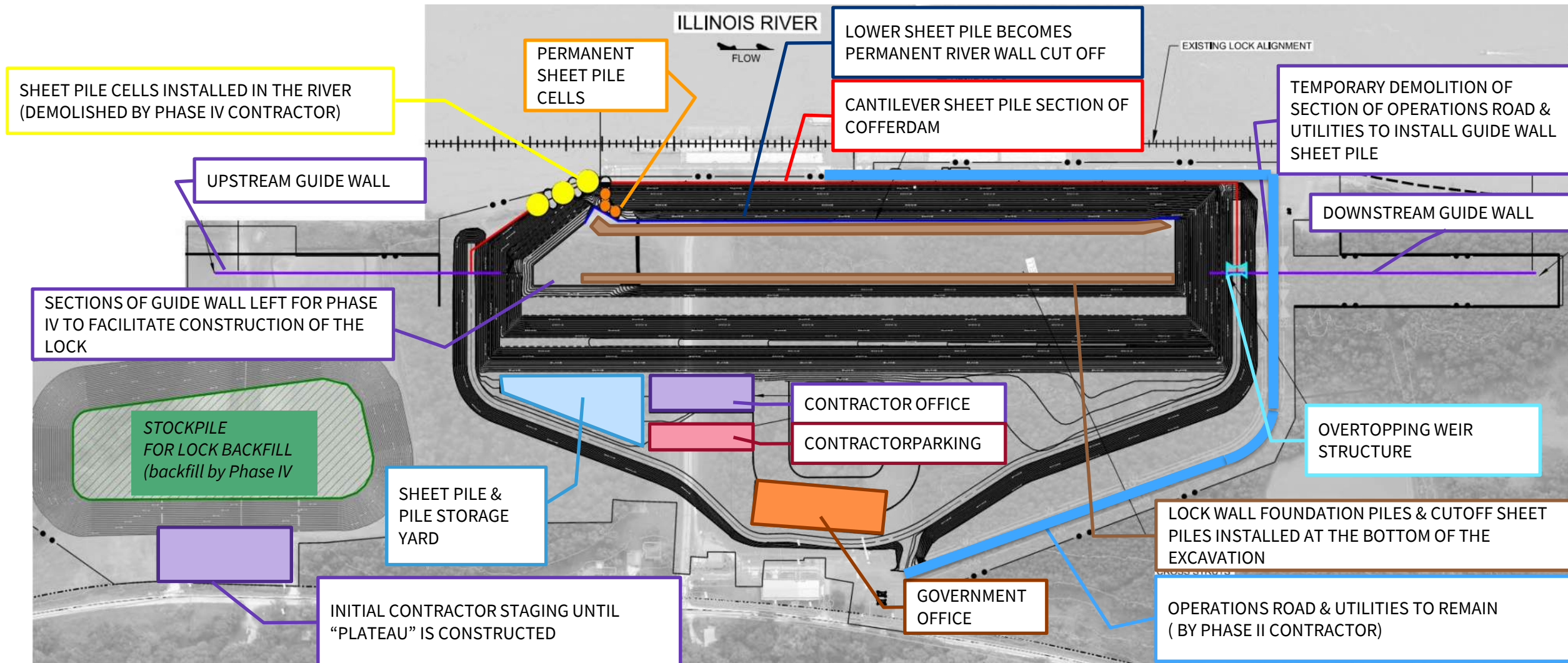


- **Total Project has 5 Phases and Construction Contracts**
 - **Phase I – HSS and Operating Machinery (UNDER CONTRACT)**
 - **Phase II – Site Development and Utilities**
 - **Phase III – Lock Excavation, Cofferdam and Foundations**
 - **Phase IV – Lock, Guide Walls, and Supporting Facilities**
 - **Phase V – Approach Channels, Site Work and Restoration**

- Clearing and Grubbing
- Operations Road
- Utilities
- Demolition of Existing Utilities

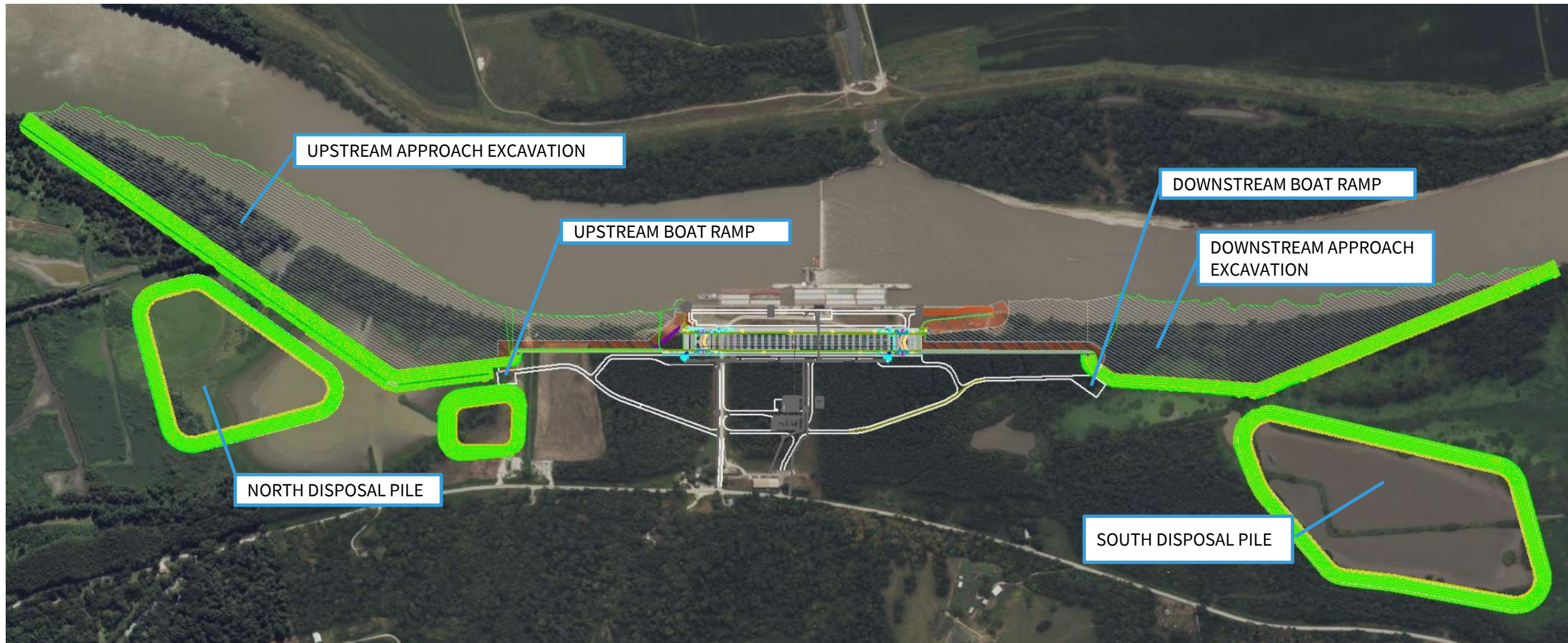


PHASE III – LOCK EXCAVATION, COFFERDAM AND FOUNDATIONS



PHASE V – APPROACH CHANNELS, SITE WORK AND RESTORATION

- Channel Approach Dredging/Excavation
- Spoil Piles
- Boat Ramps
- Final Site Restoration



CONSTRUCTION SCHEDULE

Phase I: October 2024 – October 2027 (UNDER CONTRACT)

Phase II: October 2025 – April 2026

Phase III: May 2026 – February 2029

Phase IV: February 2029 – October 2032

Phase V: November 2029 – August 2031

Contract	Cost	FY 2024	FY 2025	FY 2026	FY 2027	FY 2028	FY 2029	FY 2030	FY 2031	FY 2032
Phase I - HSS and Operating Machinery (UNDER CONTRACT)	\$10M - \$25M	■	■	■	■					
Phase II - Site Development and Utilities	\$10M - \$25M	■	■	■	■					
Phase III - Lock Excavation, Cofferdam and Foundations	\$250M - \$500M	■	■	■	■	■	■			
Phase IV - Lock, Guide Walls, and Supporting Facilities	\$500M - \$1000M	■	■	■	■		■	■	■	■
Phase V - Approach Channels, Site Work and Restoration	\$250M - \$500M	■	■	■	■			■	■	■

■ Design Fully Funded

■ Construction Funded

■ Construction Not Currently Funded

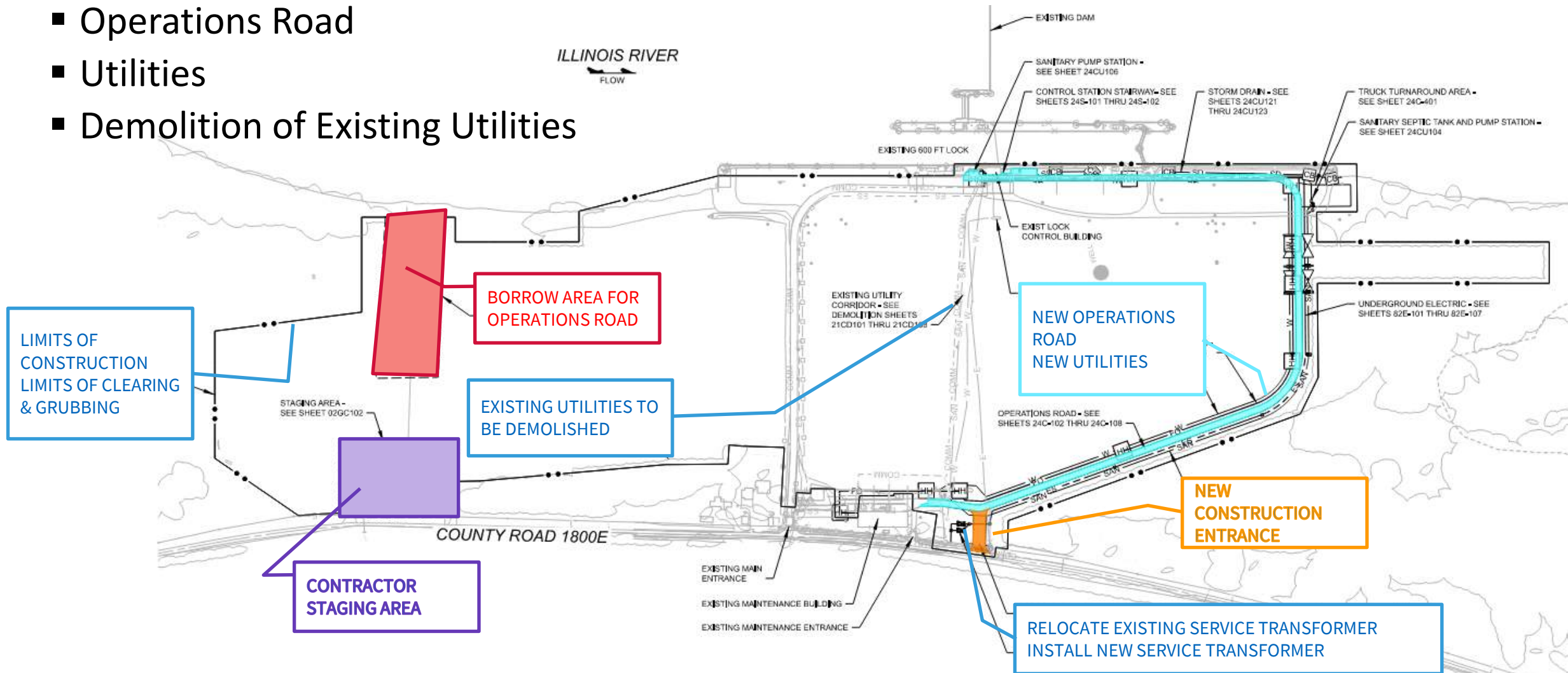
DETAILS OF PHASE I

HSS AND OPERATING MACHINERY - AWARDED

- Phase I – HSS and Operating Machinery - Awarded
 - Provided to Phase IV contractor as Government Furnished Equipment
 - HSS Gates – Similar to 600ft Lock Major Rehab
 - Miter Gates
 - Tainter Gate Culvert Valves
 - Culvert Bulkheads
 - Ice Port Gate Valves
 - Operating Machinery – Similar to 600ft Lock Major Rehab
 - HPUs
 - Miter Gate Operators and Support Frames
 - Culvert Valve Operators and Support Frame
 - Ice Port Gate Valve Operators

DETAILS OF PHASE II SITE DEVELOPMENT AND UTILITIES

- Clearing and Grubbing
- Operations Road
- Utilities
- Demolition of Existing Utilities



Environmental and Permitting Requirements

Environmental Requirements

The Supplemental Environmental Assessment prepared for this project addresses impacts to environmental resources. The following outlines agency or contractor requirements:

The USACE will be responsible for addressing the mitigation for impacted environmental resources; this is still being planned but will be in place as necessary for each contract package.

- Impacted resources include, wetlands, threatened and endangered species and their habit.

Environmental Requirements Continued...

The contractor will be required to adhere to the following:

- Removal of trees greater than 3 inches in diameter will only to be allowed between Oct 1 and March 31.
- Between the April 1 - October 31 timeframe, work zone lighting must be directed downward and pointed away from forested areas to minimize disturbance to protected bat species at night.
- Seeds from the Decurrent False Aster plants will be collected by USACE staff in the fall of 2025. Areas where these plants are growing within the project limits and from which seed is planned to be collected will be marked as no intrusion areas on the plans.
- No Eagle nests have been found with the project limits. As work progresses, should one be discovered within 660 feet of the construction zone, the work will stop until further coordination with the UW Fish and Wildlife Service has occurred.

Environmental Requirements Continued...

The contractor will be required to adhere to the following:

- Previously used equipment and watercraft will be required to be cleaned prior to bringing it onto or removing it from the site to prevent the spread of invasive species.
- Significant historical sites have been identified within the project limits. A Programmatic Agreement between USACE, Illinois SHPO, the Miami Tribe of Oklahoma, and the Osage Nation has been developed. This agreement documents how identified sensitive areas will be addressed before, during, and after construction.
- If petroleum impacted soil or groundwater is encountered during construction, further investigation and evaluation may be required; contaminated materials will be properly handled and disposed.

Environmental Requirements Continued...

The contractor will be required to adhere to the following:

- Conduct work during winter months, as able, to reduce impacts to navigation.
- Federal, state, and local air resource requirements to minimize impacts to air quality.
- Illinois Pollutant Discharge Elimination permit and accompanying Stormwater Pollution Prevention Plan
- Section 401 Water Quality Certification issued by the ILDNR.
- Excavated material will be available for free public beneficial use.

Permitting Requirements

The following permits will be handled by USACE staff. Contract specifications will include and conditions that must be followed.

- Section 10 and 404, verified for compliance by USACE
- Section 401, filed by USACE
- Floodway Permit, filed by USACE
- Dam Permit Modification, filed by USACE
- LOMR and CLOMR, being determined, to be filed by USACE
- County Road Entrance Permit, filed by USACE

The following permits will be handled by the Contractor

- NPDES Construction Permit

DETAILS OF PHASE II SITE DEVELOPMENT AND UTILITIES

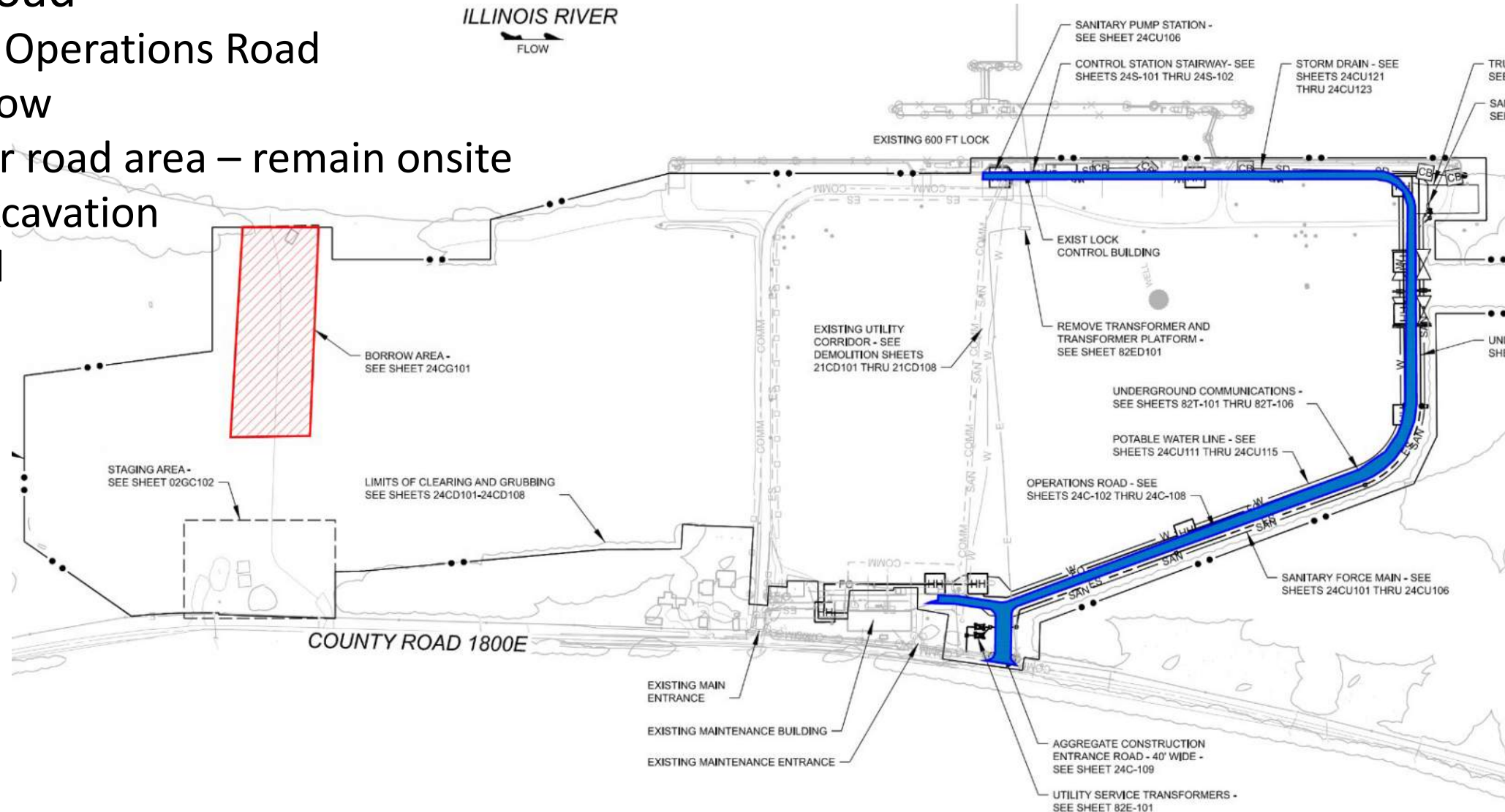
- Clearing and Grubbing
 - Majority of clearing to be completed in this phase (42 Acres)
 - Clearing between October 1st and March 31st only
 - Preparing site for grading and stockpile areas in Phase III and IV



DETAILS OF PHASE II SITE DEVELOPMENT AND UTILITIES

■ Operations Road

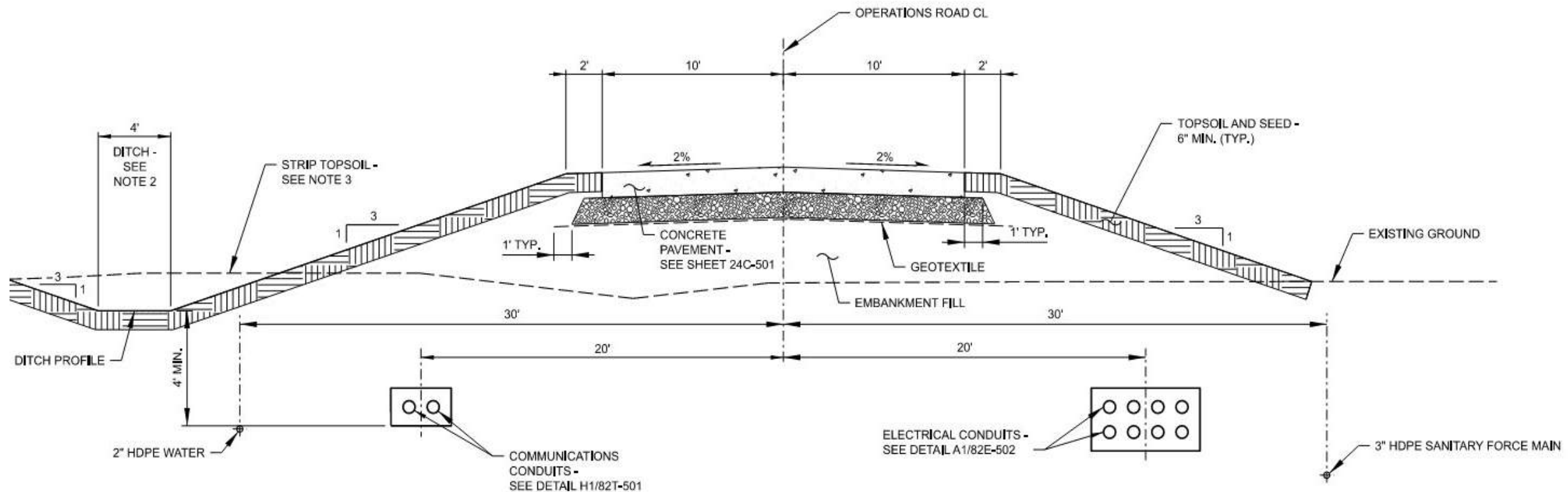
- Grading for Operations Road
- Onsite borrow
- Stripping for road area – remain onsite
- 1,700 cy excavation
- 8,200 cy fill



DETAILS OF PHASE II SITE DEVELOPMENT AND UTILITIES

■ Operations Road

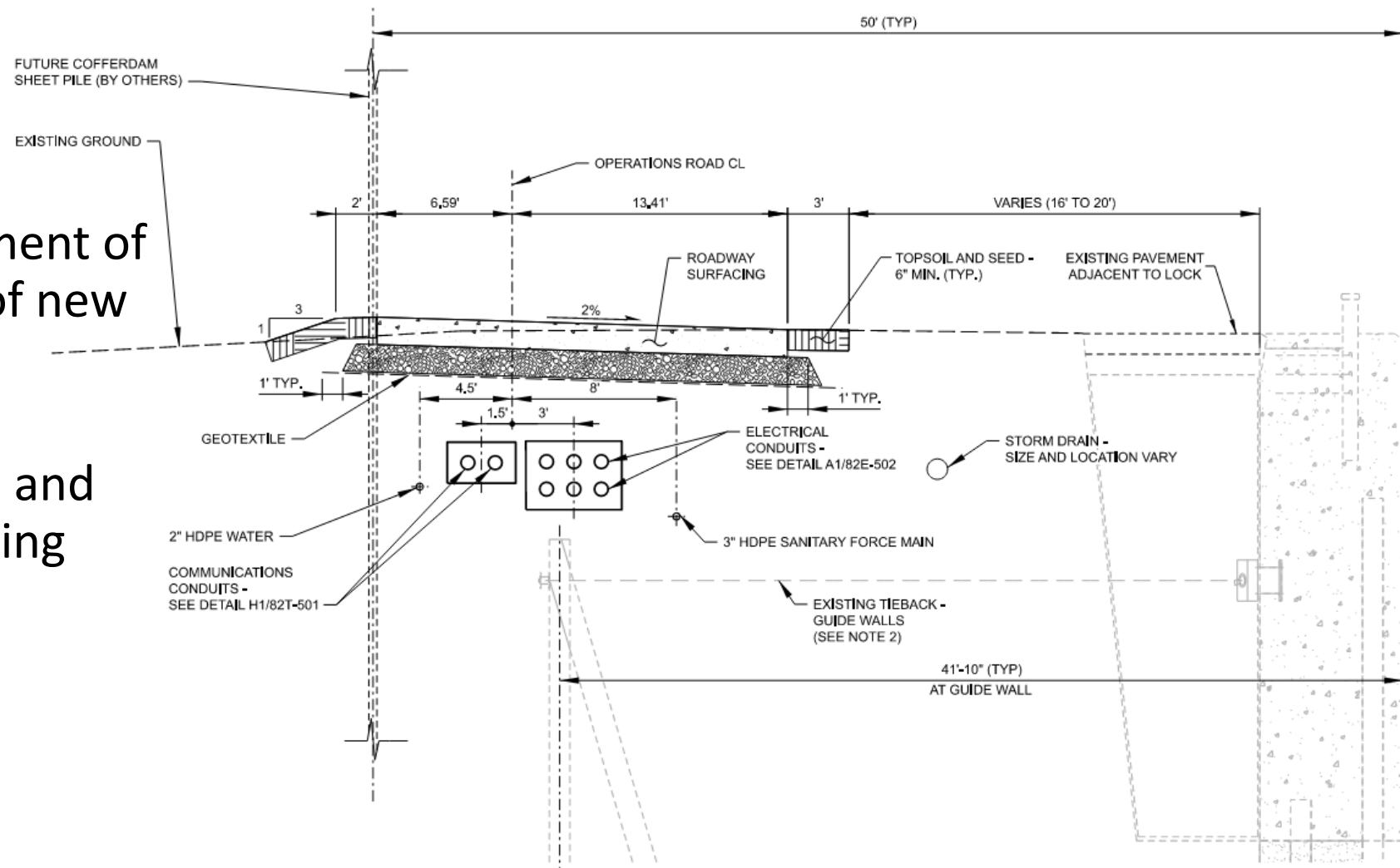
- Paving: concrete 8" thick pavement
- Concrete surface for entire length



DETAILS OF PHASE II SITE DEVELOPMENT AND UTILITIES

■ Utilities

- Sanitary, Water, Storm
- Electric
- Telecommunication
- Demolition and abandonment of existing after installation of new lines
- Congested utility corridor between existing lock wall and future cofferdam sheet piling

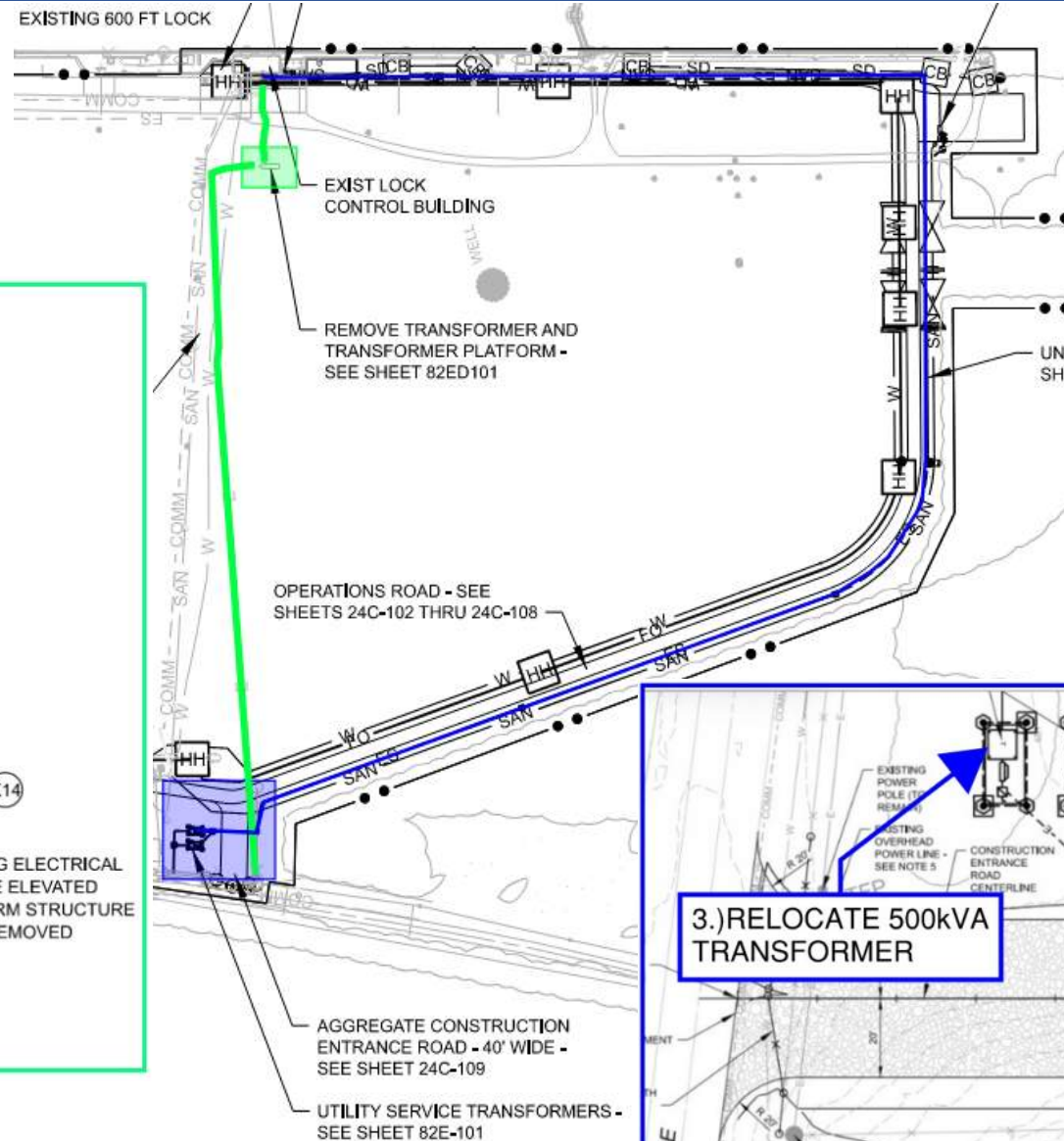
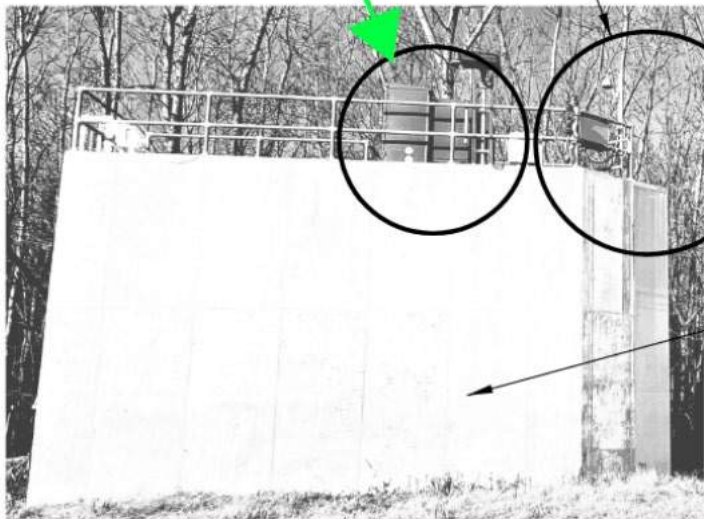


DETAILS OF PHASE II SITE DEVELOPMENT AND UTILITIES

■ Utilities

- Relocate existing 500kVA transformer
- Provide new 1500kVA transformer

2.) REMOVE EXISTING 500kVA TRANSFORMER



3.) RELOCATE 500kVA TRANSFORMER

1.) NEW 1500kVA TRANSFORMER

DETAILS OF PHASE II

ESTIMATED CONSTRUCTION LOADS

ESTIMATED SITE POWER REQUIREMENTS BY PHASE

	GOVERNMENT FACILITIES						CONSTRUCTION CONTRACTOR					
	Estimated Peak Demand (KVA)	New Loads (kVA)	Source from Service Transformer During Phase 2	Source from Service Transformer During Phase 3	Source from Service Transformer During Phase 4	Load Description	Estimated Peak Demand (KVA)	New Loads (kVA)	Source from Service Transformer During Phase 2	Source from Service Transformer During Phase 3	Source from Service Transformer During Phase 4	Load Description
Existing 600' Lock Phase 1	250	0	[2] then [1]*	[1]*	[2]*	Peak Existing Load	0	0	N/A	N/A	N/A	
Contract Package Phase 2 - Site Clearing and temporary power to existing lock	250	0	N/A	N/A	N/A	Procurement Phase	0	0	N/A	N/A	N/A	Procurement Phase
Contract Package Phase 3 - Excavation and foundation	250	0	N/A	N/A	N/A	No new loads	1350	68	*	*	*	Contractor Trailers (2x 24KVA) & USACE single wide (20 kVA) Temporary Yard Illumination
								150	N/A	[2] OR *	*	3 x double-wide trailers (150 KVA) - construction contractor
								120	N/A	[2] OR *	*	Mechanic Shop & yard illumination (120KVA) - construction contractor
								50	N/A	[2] OR *	[2] OR *	USACE Office double-wide trailer (50KVA temp)
								7	N/A	[2] OR *	[2] OR *	Well Water Pump (5HP)
								45	N/A	[2] OR *	[2] OR *	Pumps for Guide walls/Sumps/other systems (20 HP x 2)
								1,000	N/A	[2]	[2]	Main Lock excavation - Dewatering & Deep wells pumps (~60hp total multiple pumps) Preliminary estimate 800 - 1000 HP possible.
Contract Package Phase 4 - Lock Construction (With temporary USACE power still active)	1000						2550	350	N/A	[2]	[2]	Add six more USACE Contract Work double-wide trailers, QA Mat Lab, and parking lights. (6x 50KVA)
								1,000	N/A	N/A	[3]	Concrete Batch Plant (750kVA)
								200	N/A	N/A	[3]	Central Air Compressor (200KVA)

■ Utilities

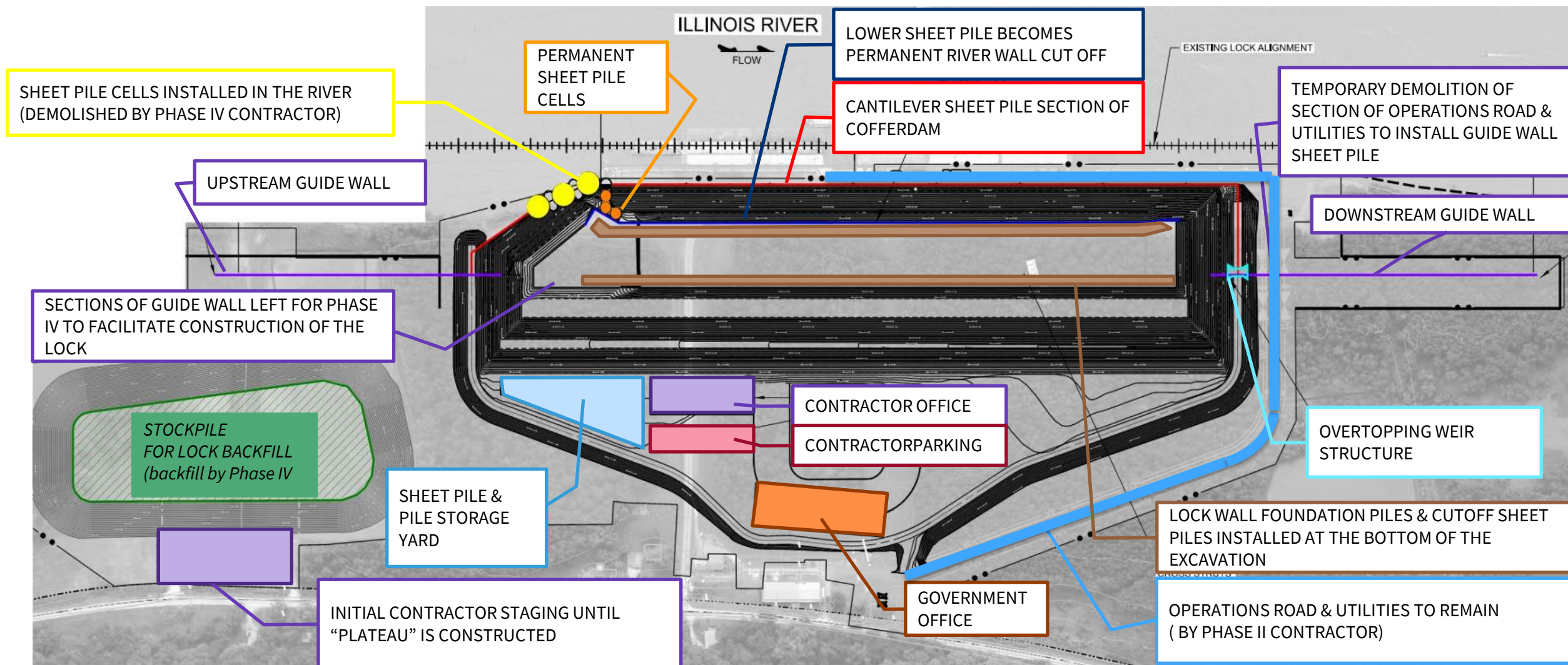
- Keep existing 600' Lock operational loads and metering separate from construction loads.
- 1500kVA transformer to be new utility transformer for new 1200' lock distribution system.
- Third transformer required for construction loads to run concurrently with site distribution in Phase 4.

■ Construction Electrical Loads

- Loads are estimated. Feedback on estimated loads

DETAILS OF PHASE III

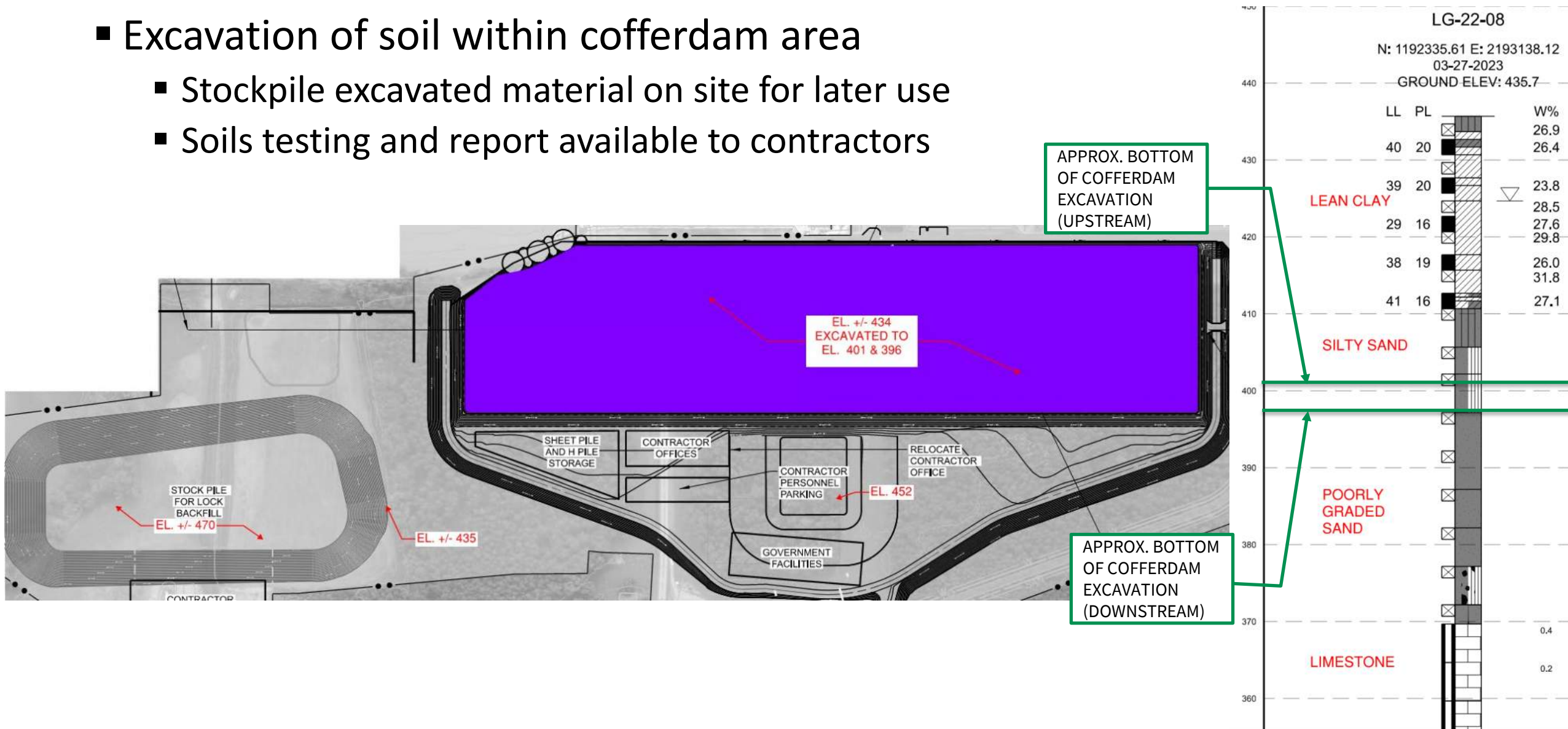
LOCK EXCAVATION, COFFERDAM AND FOUNDATIONS



DETAILS OF PHASE III

LOCK EXCAVATION, COFFERDAM AND FOUNDATIONS

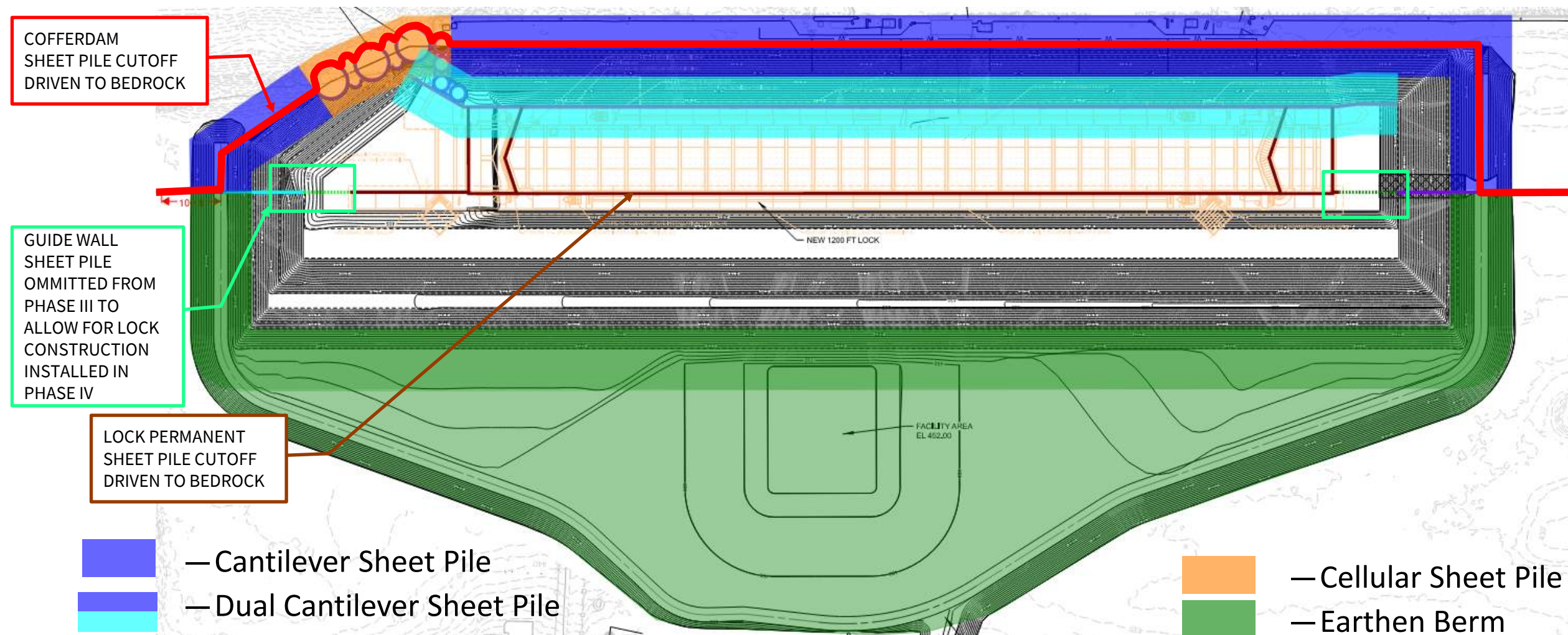
- Excavation of soil within cofferdam area
 - Stockpile excavated material on site for later use
 - Soils testing and report available to contractors



DETAILS OF PHASE III

LOCK EXCAVATION, COFFERDAM AND FOUNDATIONS

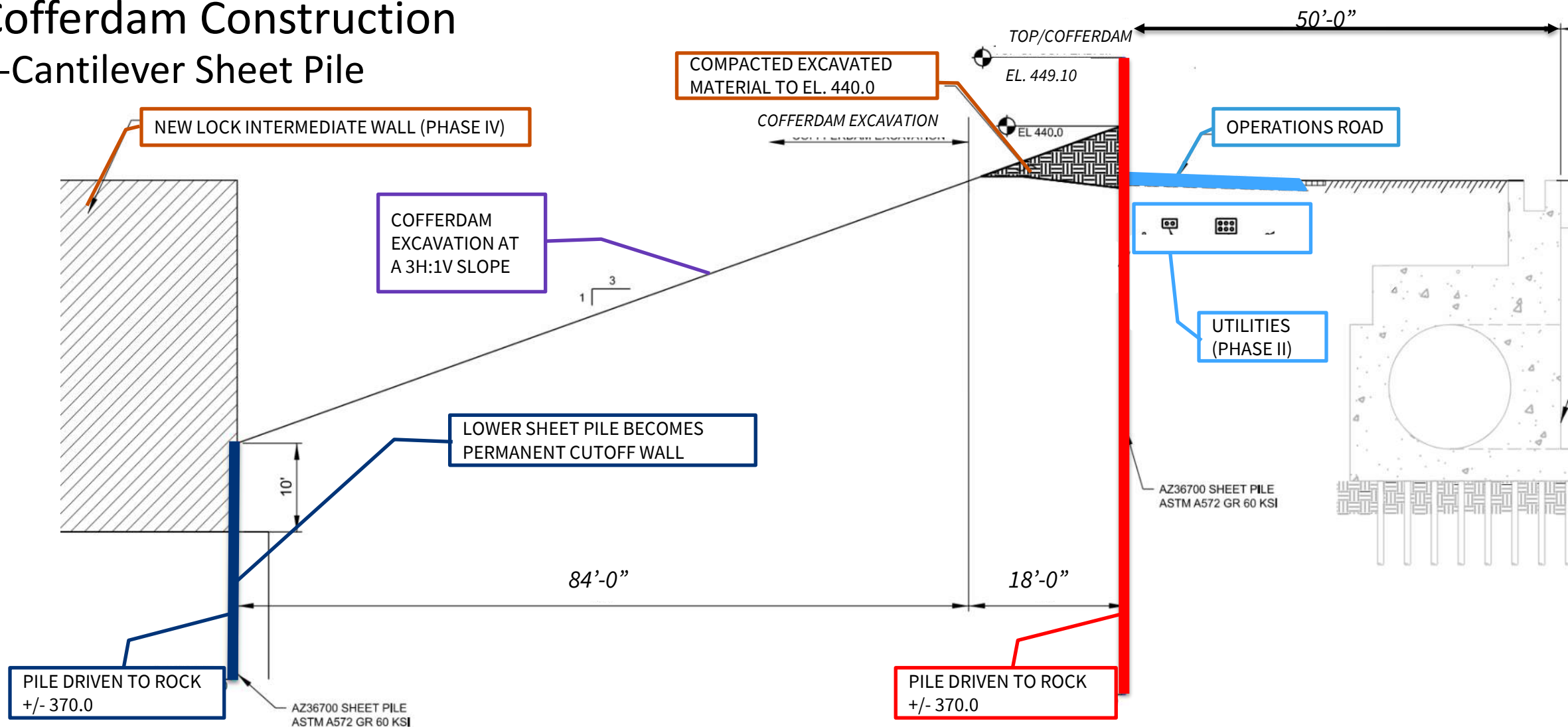
■ Cofferdam Construction



DETAILS OF PHASE III

LOCK EXCAVATION, COFFERDAM AND FOUNDATIONS

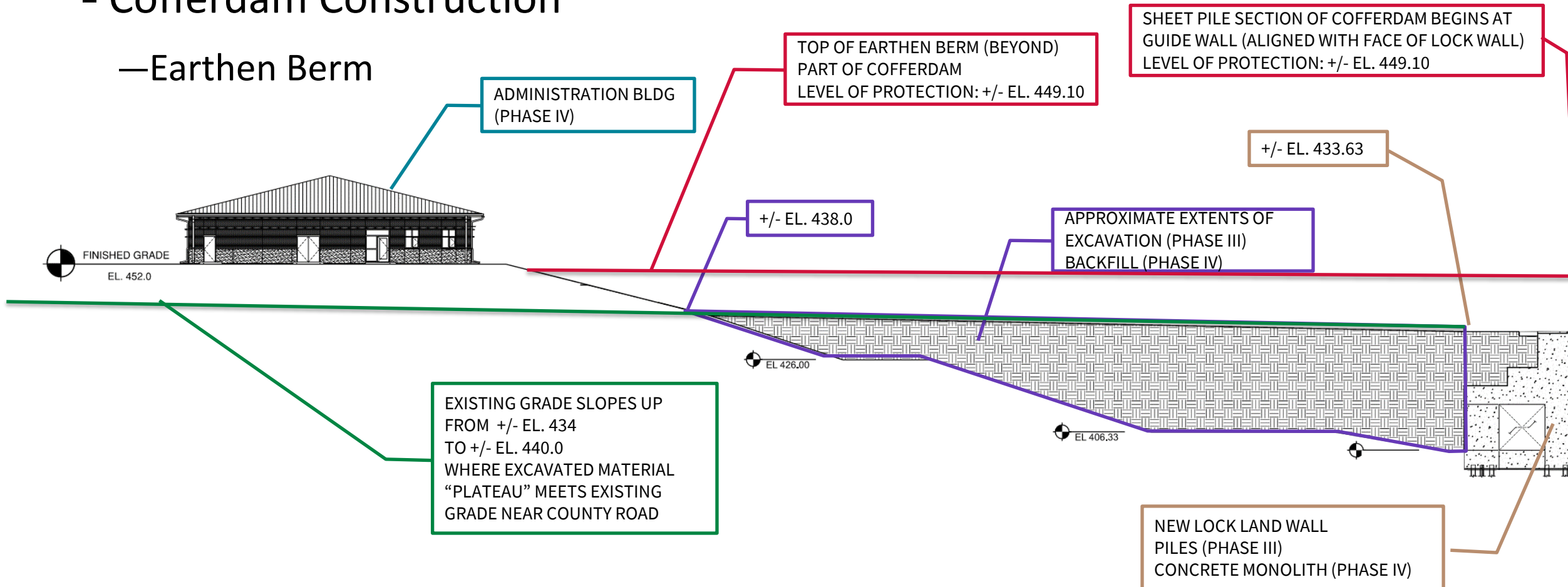
■ Cofferdam Construction —Cantilever Sheet Pile



DETAILS OF PHASE III LOCK EXCAVATION, COFFERDAM AND FOUNDATIONS

■ Cofferdam Construction

—Earthen Berm



DETAILS OF PHASE III

LOCK EXCAVATION, COFFERDAM AND FOUNDATIONS

■ Dewatering System

— Consists of

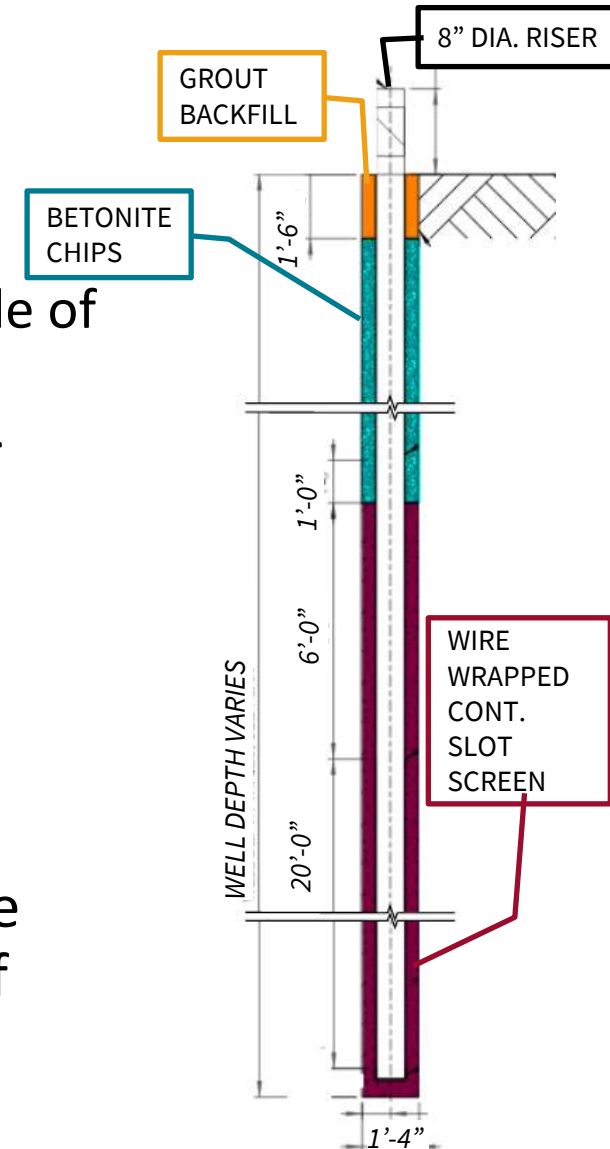
- Sheet pile driven to bedrock
- A series of approximately 20 well points along the chamber side of the new lock intermediate wall & the benched land side slope

—Final design of dewatering system by the Phase III Contractor

—Electrical service for dewatering system pumps by Phase II Contractor

—Operation & maintenance of the dewatering system to be transferred to the Phase IV Contractor as part of the Phase III Close Out

—Operation & maintenance of the dewatering system to be the responsibility of the Phase IV Contractor for the remainder of the lock construction

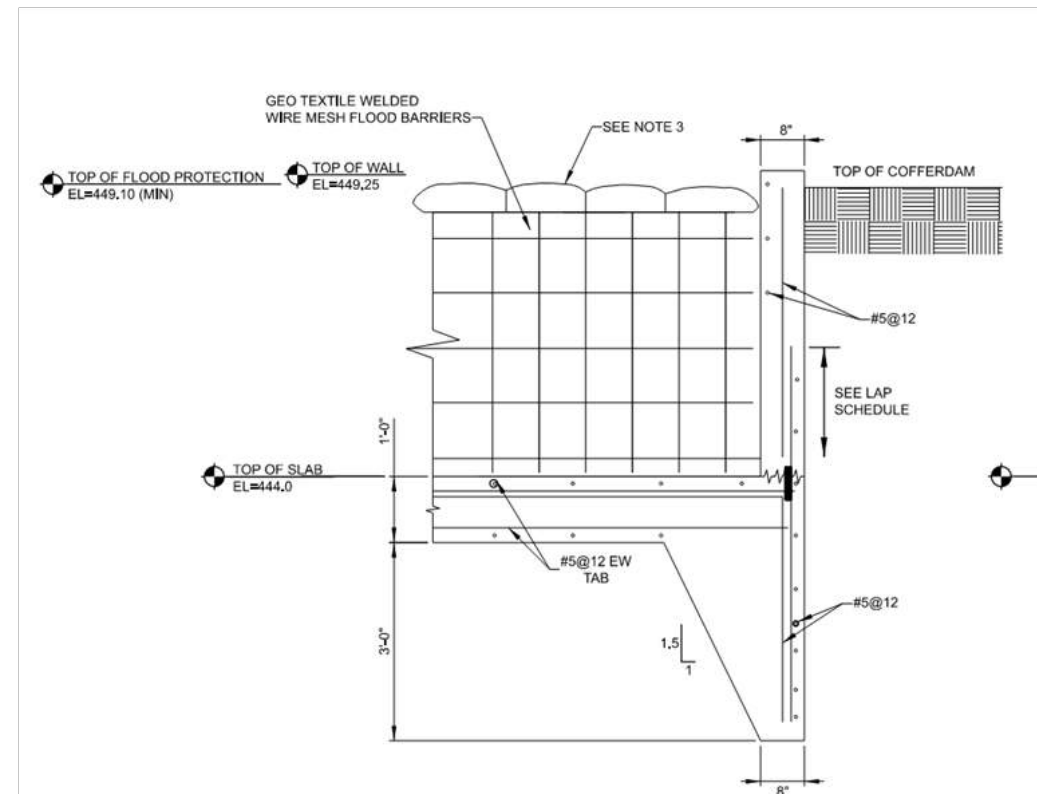
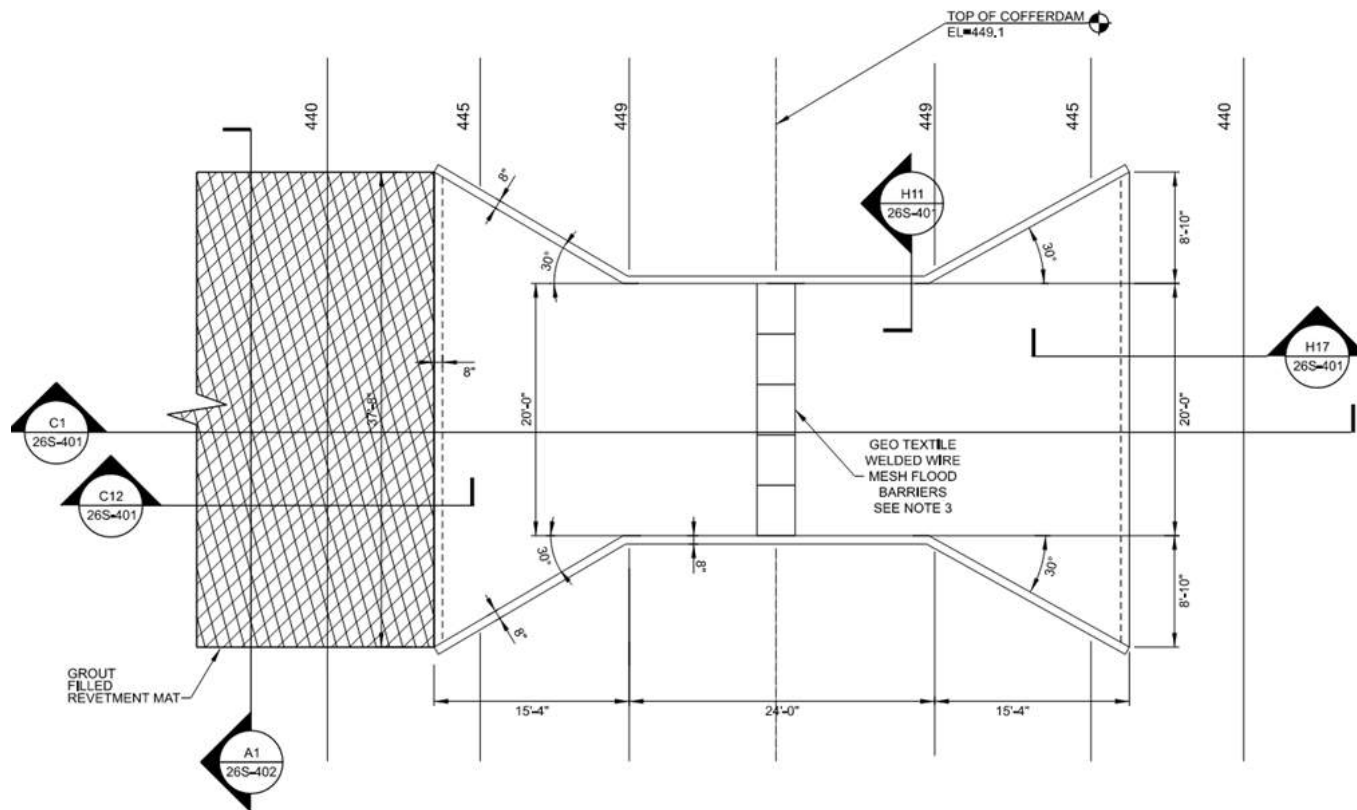


DETAILS OF PHASE III

LOCK EXCAVATION, COFFERDAM AND FOUNDATIONS

■ Overtopping Control Weir

—Flood barriers are removed to allow the cofferdam to flood before an overtopping event

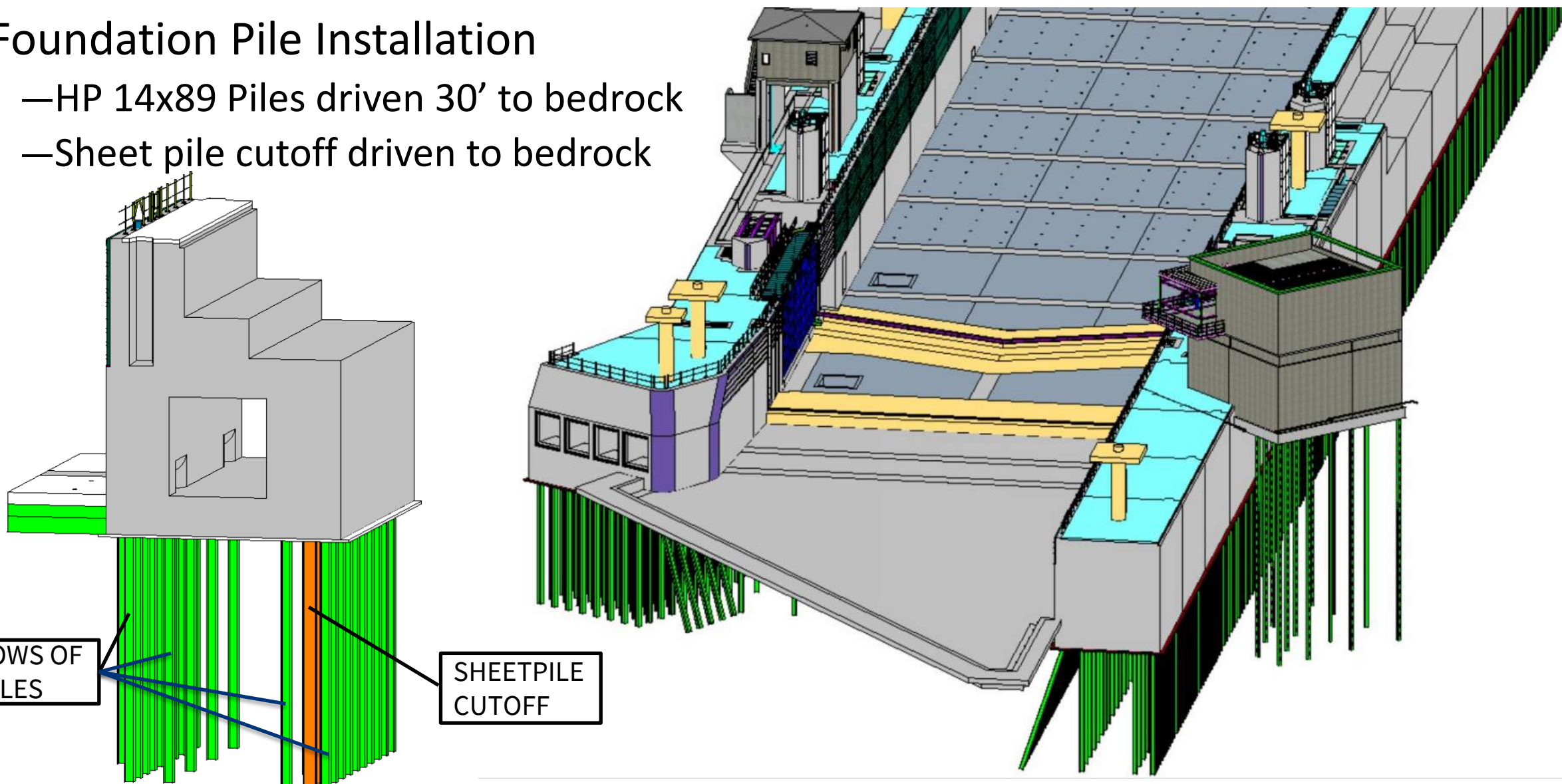


DETAILS OF PHASE III

LOCK EXCAVATION, COFFERDAM AND FOUNDATIONS

■ Foundation Pile Installation

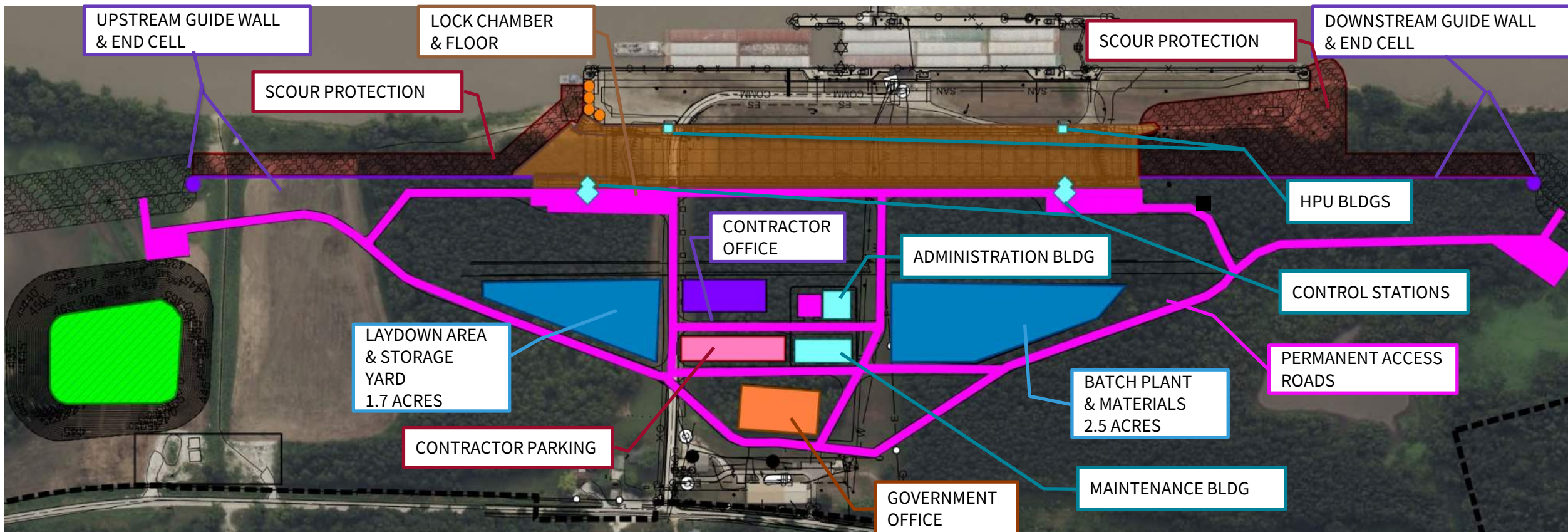
- HP 14x89 Piles driven 30' to bedrock
- Sheet pile cutoff driven to bedrock



DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

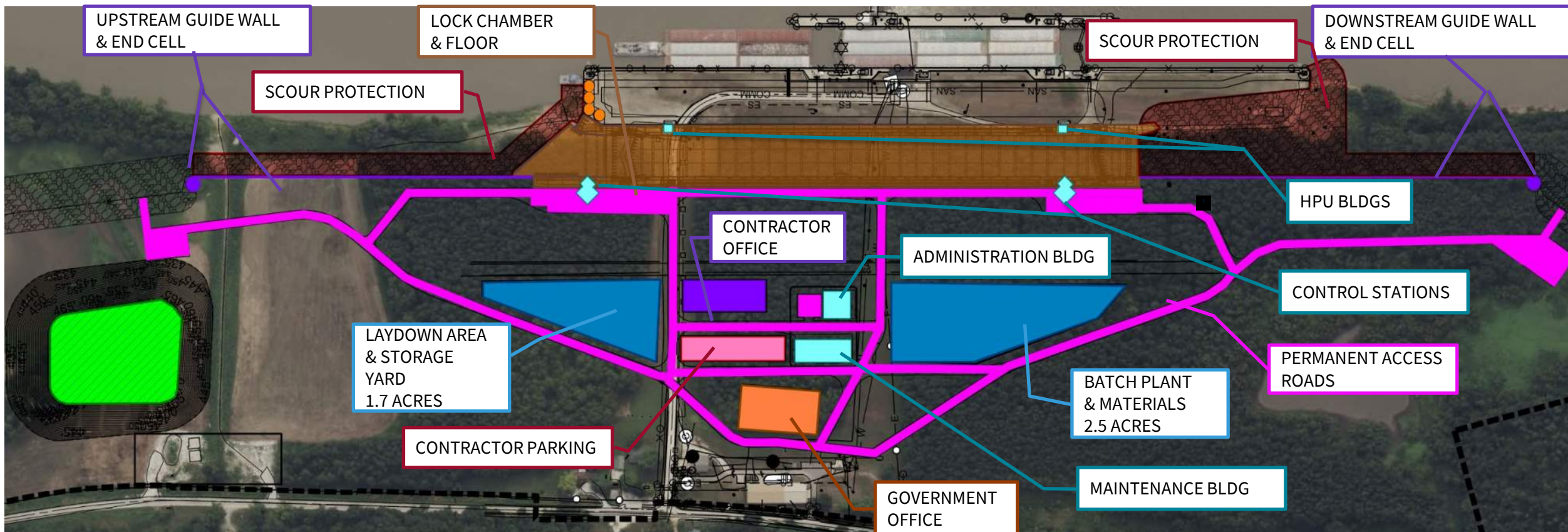
- Batch Plant & Contractor Facilities
- Lock Chamber Concrete
- Installation of Steel Fabrications from Phase I
- Guide Walls & End Cells



DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

- Supporting Buildings
- Controls, Communications, and Monitoring
- Excavation & Dismantling of the Cofferdam



DETAILS OF PHASE IV

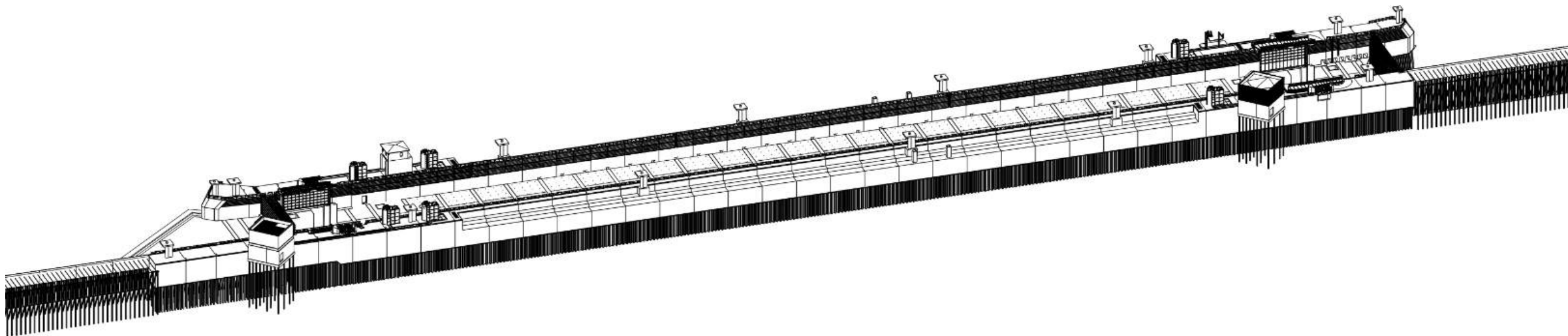
LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

- Batch Plant & Contractor Facilities
 - Batch Plant vs. Redimix
 - Power
 - Service transformers installed in Phase II
 - Site Distribution Estimated Loads – 1 MVA
 - Construction (Contractors) estimated Loads:
 - Concrete Batch Plant - 1MVA.
 - Central Air Compressor - 200kVA.
 - Water /tanks
 - Ice plant
 - Available space for laydown areas & staging
 - Access
 - Unloading Pier
 - Other anticipated needed facilities?

DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

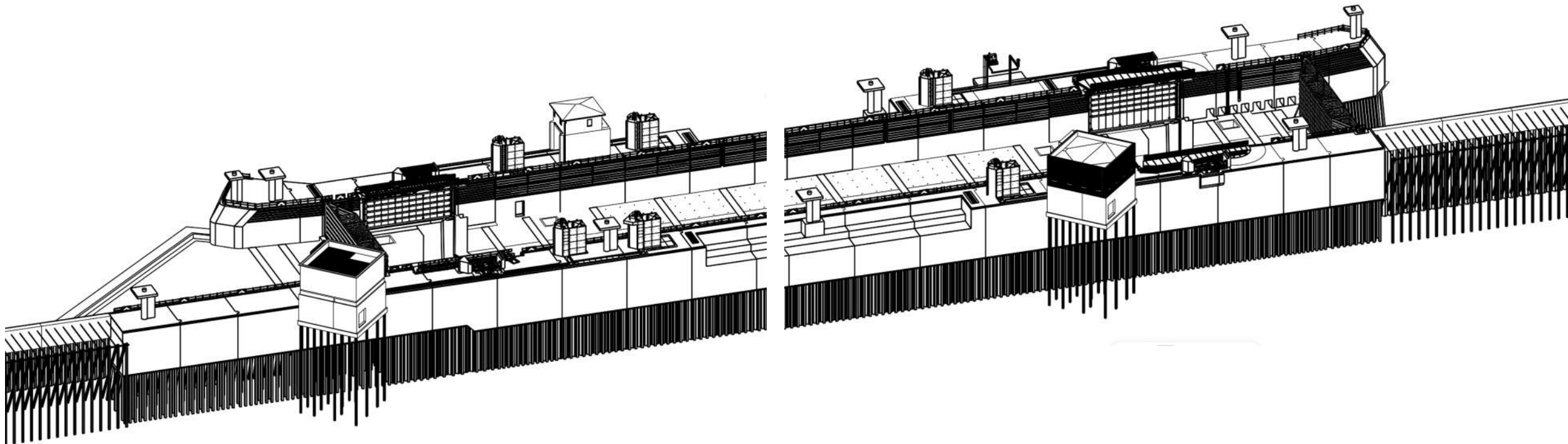
- Lock Chamber Concrete
 - Lock Wall Monoliths
 - Lock Floor Features (sills, floor struts, lock floor)
 - Debris Protection Structures



DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

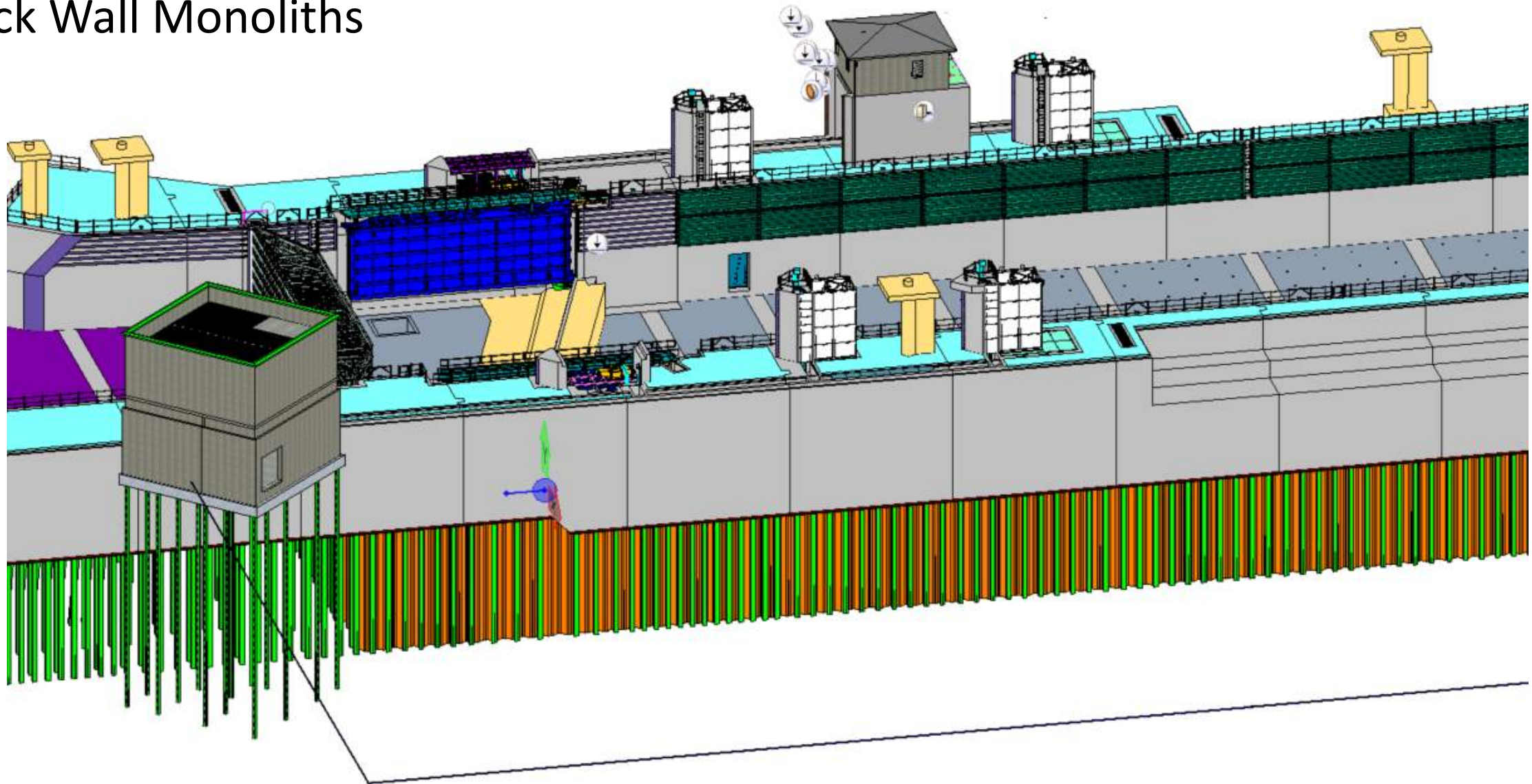
- Lock Wall Monoliths
 - CIP Concrete
 - 12'-6" Square Fill-Empty Culvert
 - HP-Pile Foundation – Piles Founded On Bedrock



DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

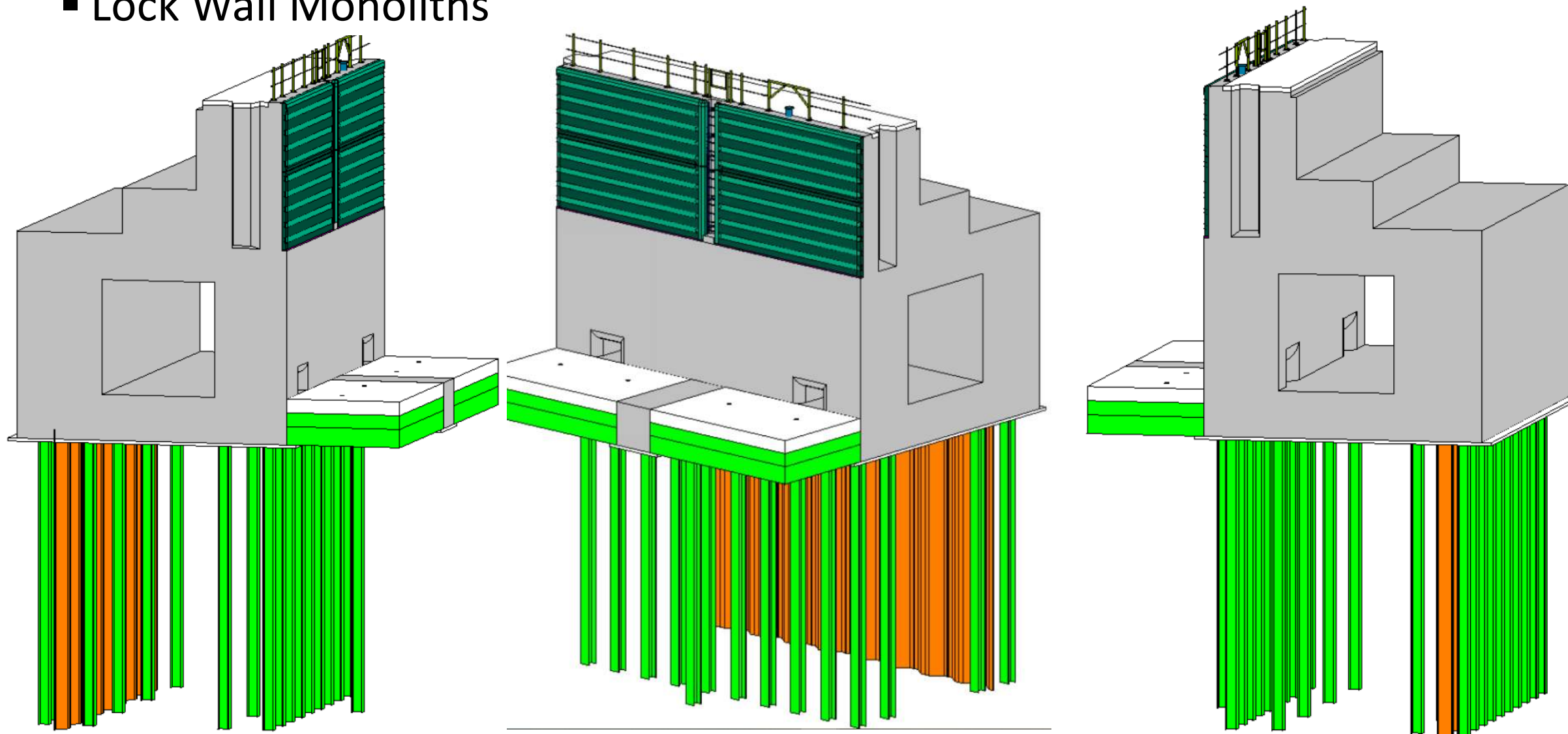
- Lock Wall Monoliths



DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

■ Lock Wall Monoliths

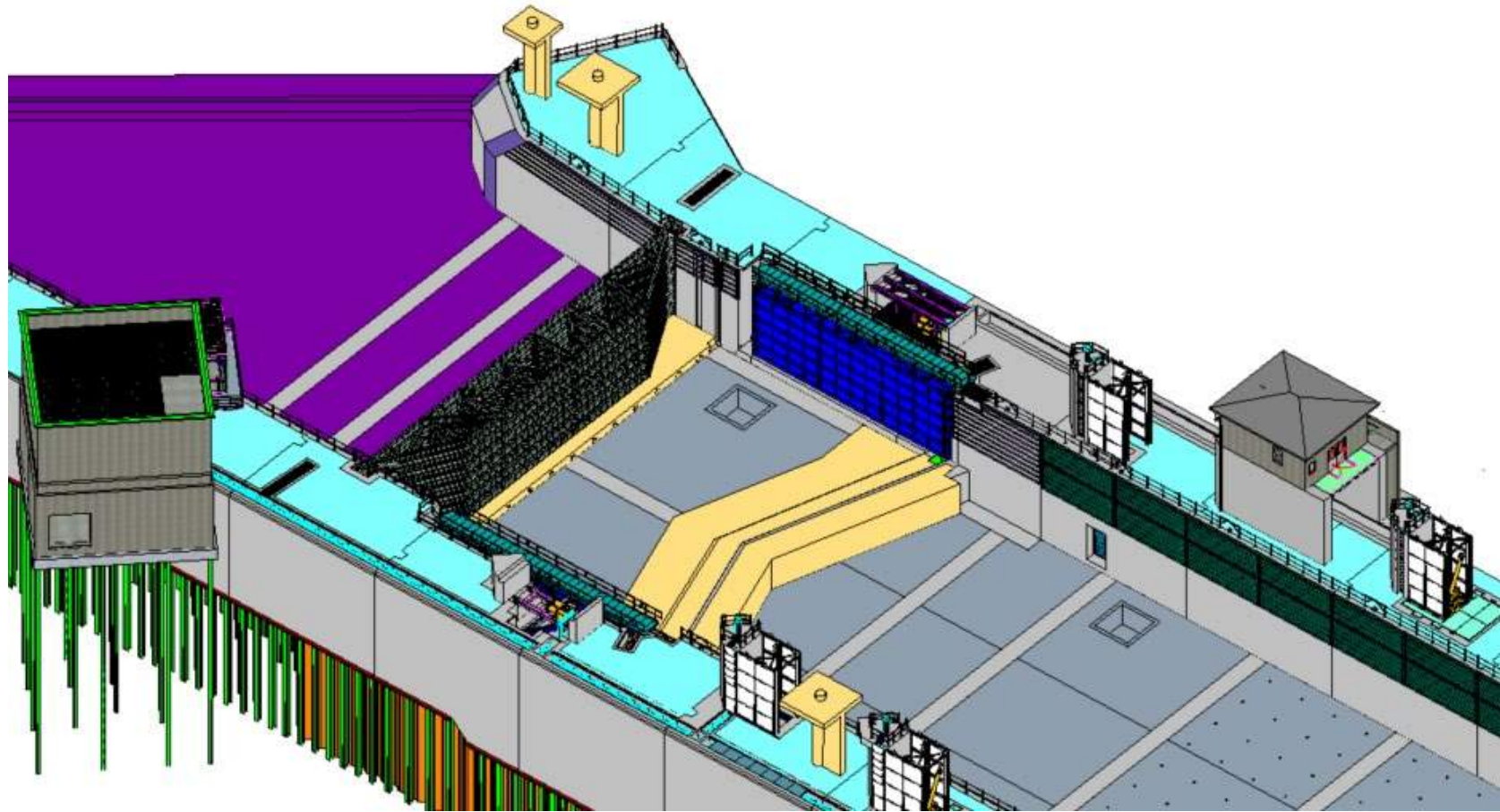


DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

■ Lock Floor Features

- CIP Concrete Bulkhead Sills
- CIP Concrete Miter Gate Sills
- CIP Concrete Lock Floor Struts
- CIP Concrete Lock Floor

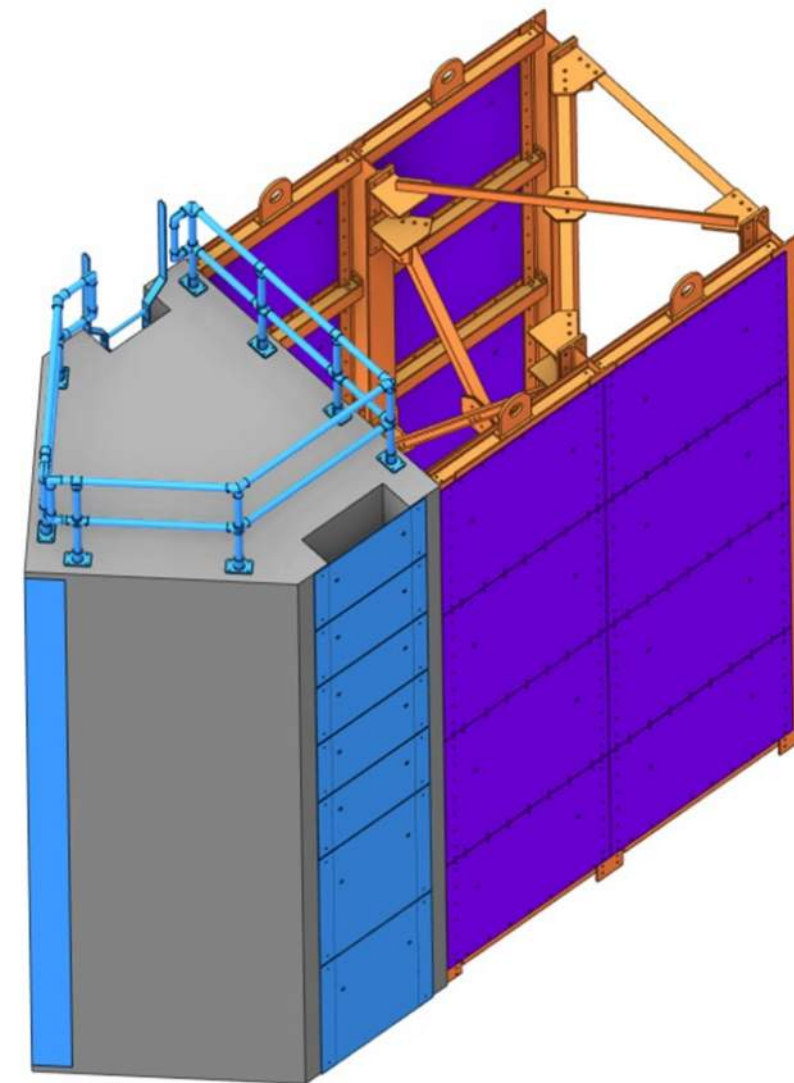
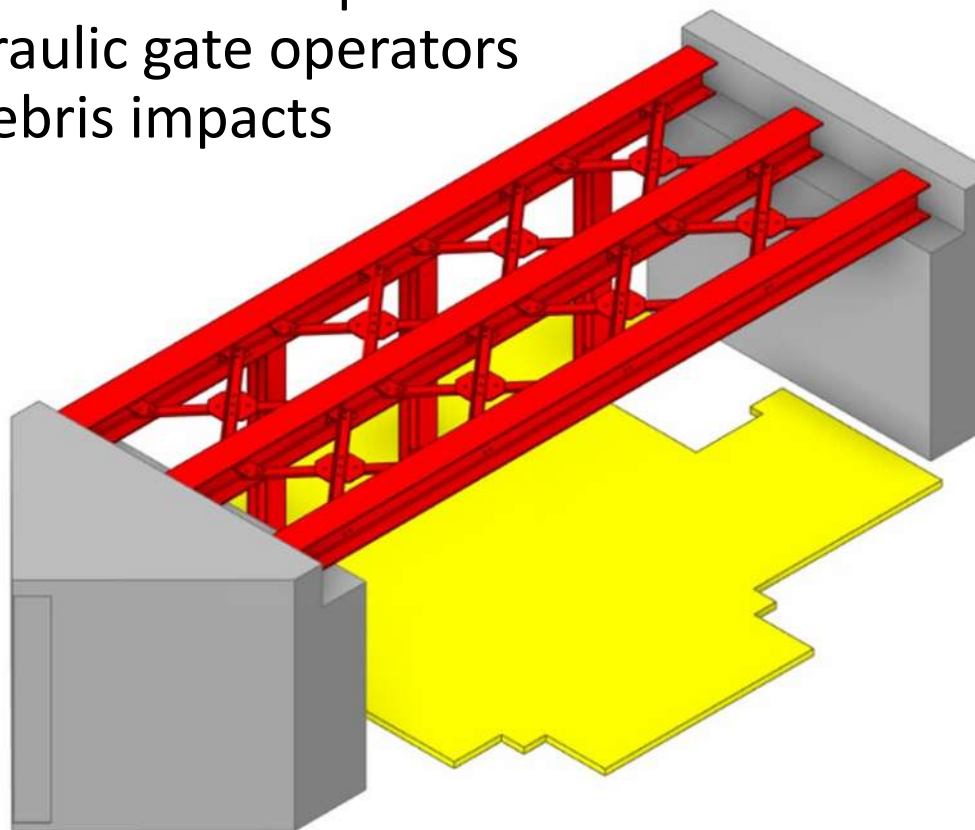


DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

■ Debris Protection Structures

- Lock Site Floods during Winter Closure
- Protection piers and frames prevent damage to hydraulic gate operators from floating debris impacts



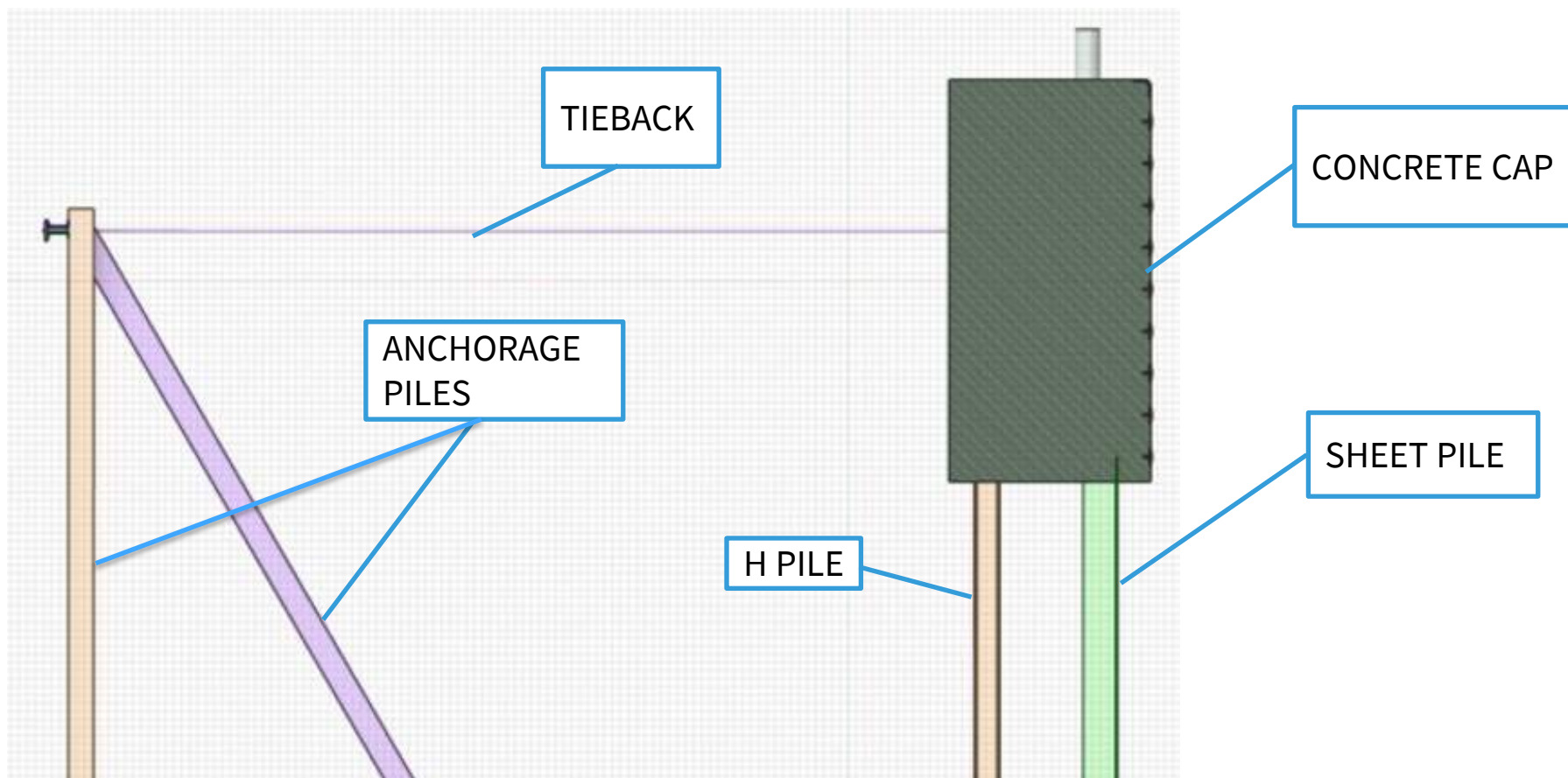
DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

■ Guide Walls

—Downstream Guide Wall = 1056 ft long

-Upstream Guide Wall = 912 ft long

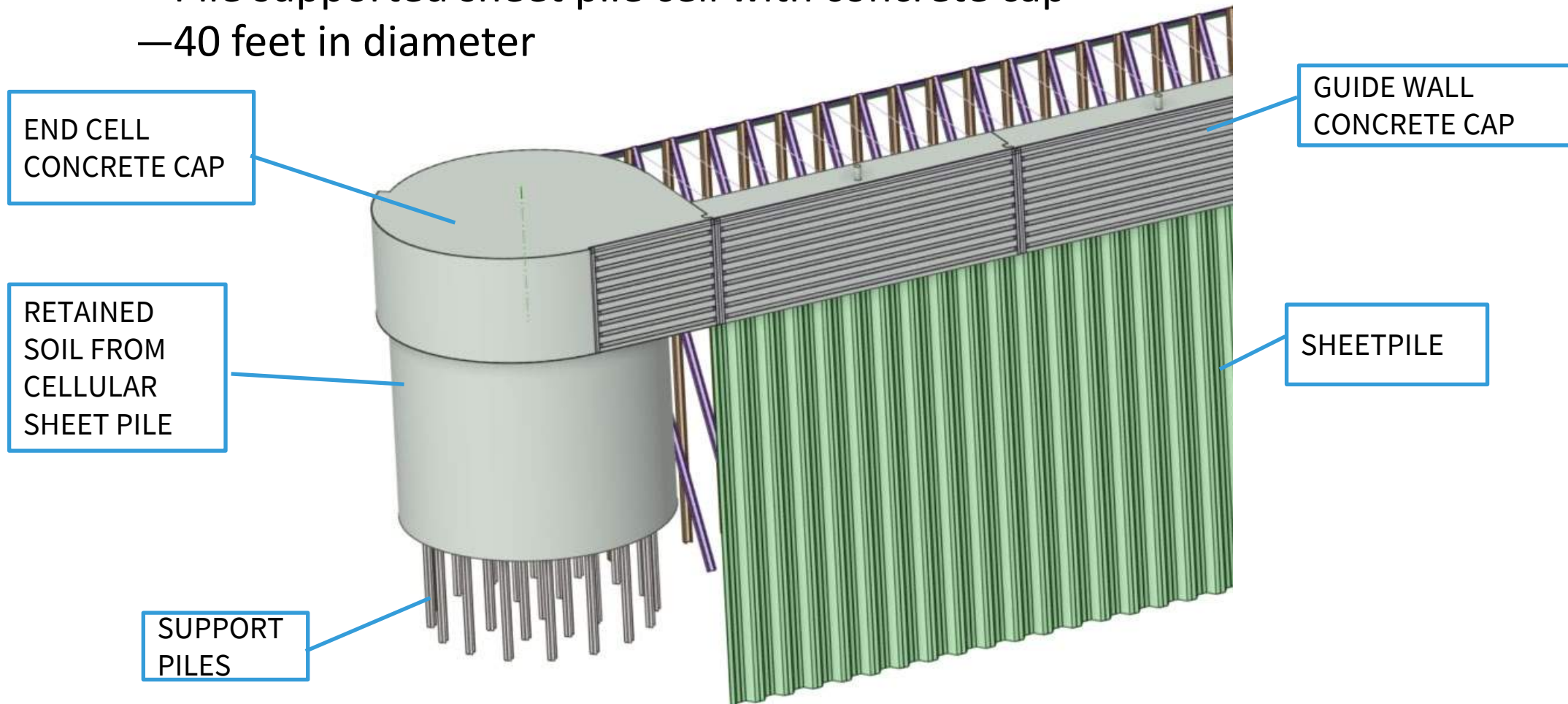


DETAILS OF PHASE IV IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

■ End Cells

- Pile supported sheet pile cell with concrete cap
- 40 feet in diameter



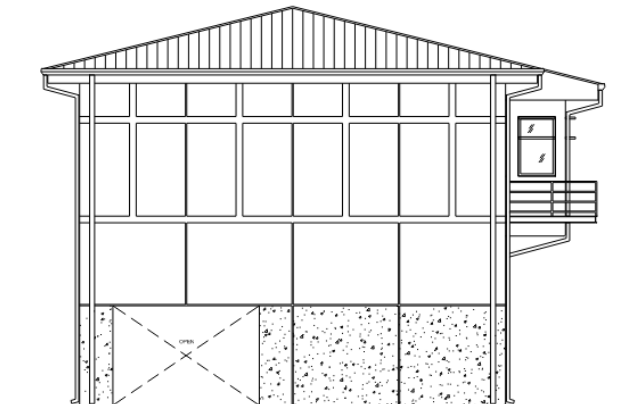
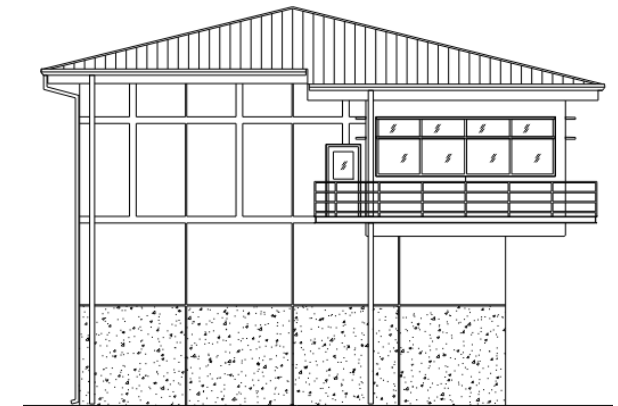
DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

- Supporting Buildings
 - Upstream & Downstream Control Stations
 - Maintenance Building



Maintenance Building Elevations

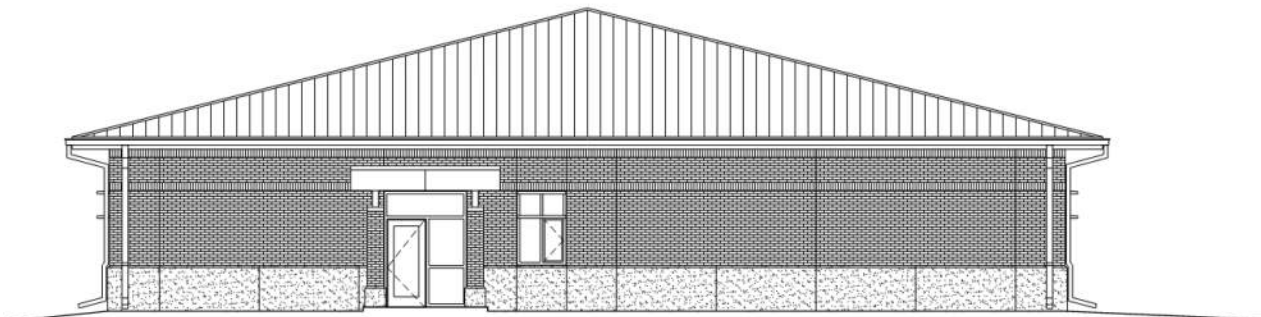


Control Station Elevations

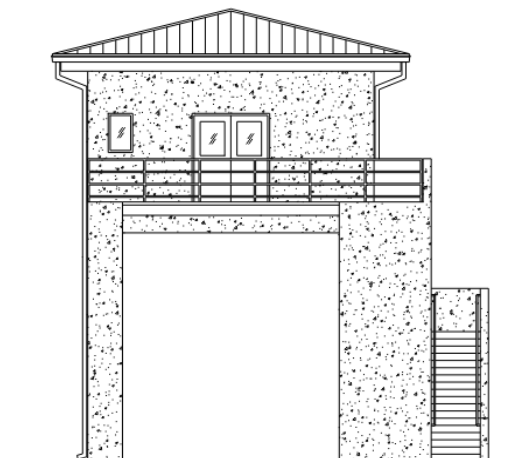
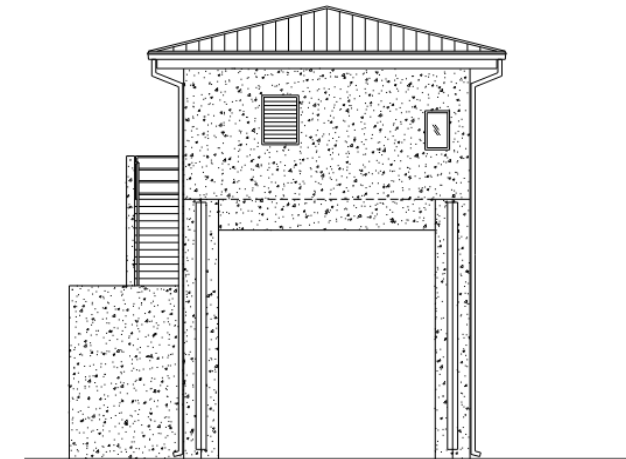
DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

- Supporting Buildings
 - Upstream & Downstream HPU Buildings
 - Administration Building



Administration Building Elevations

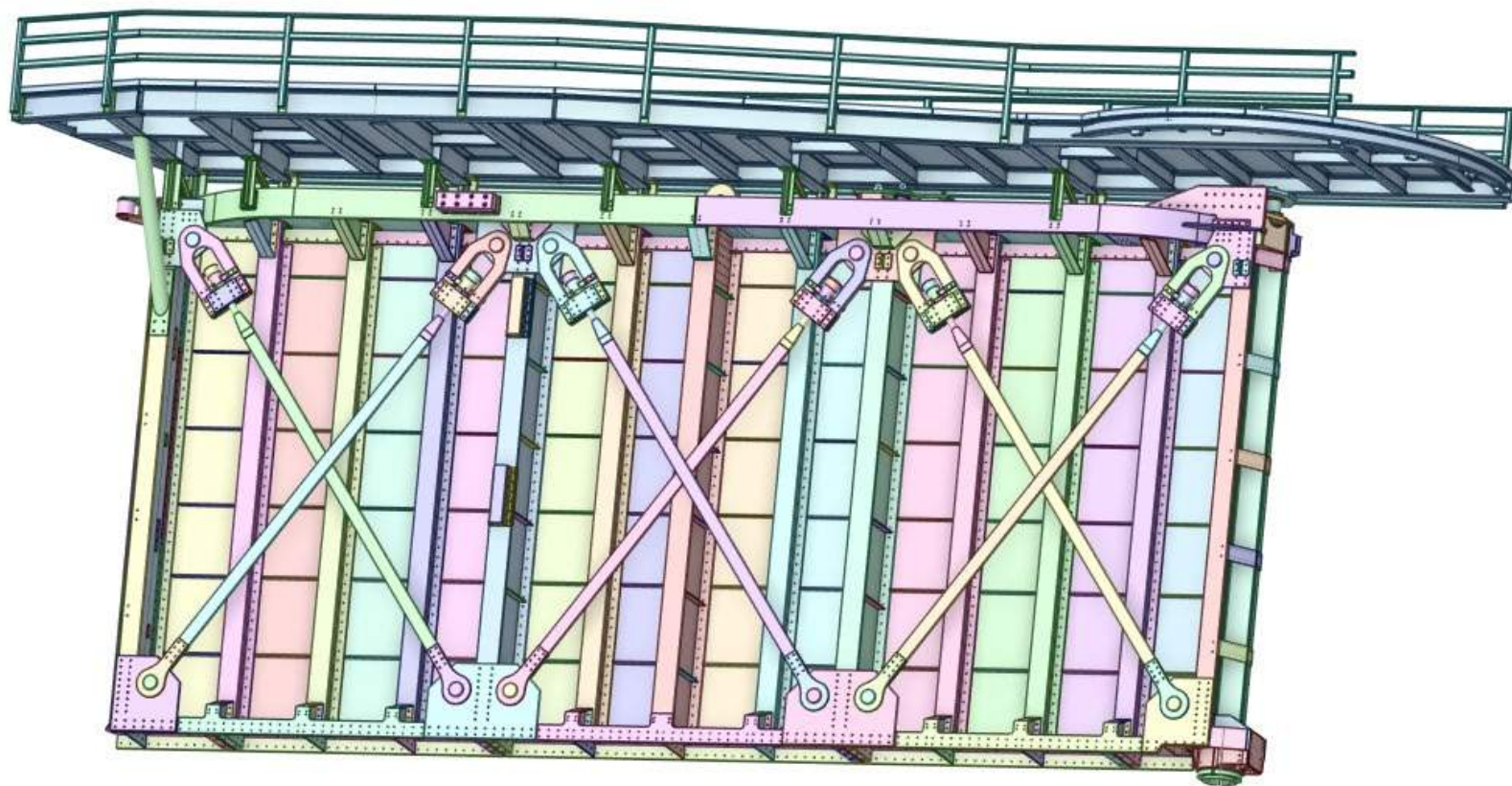


HPU Building Elevations

DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

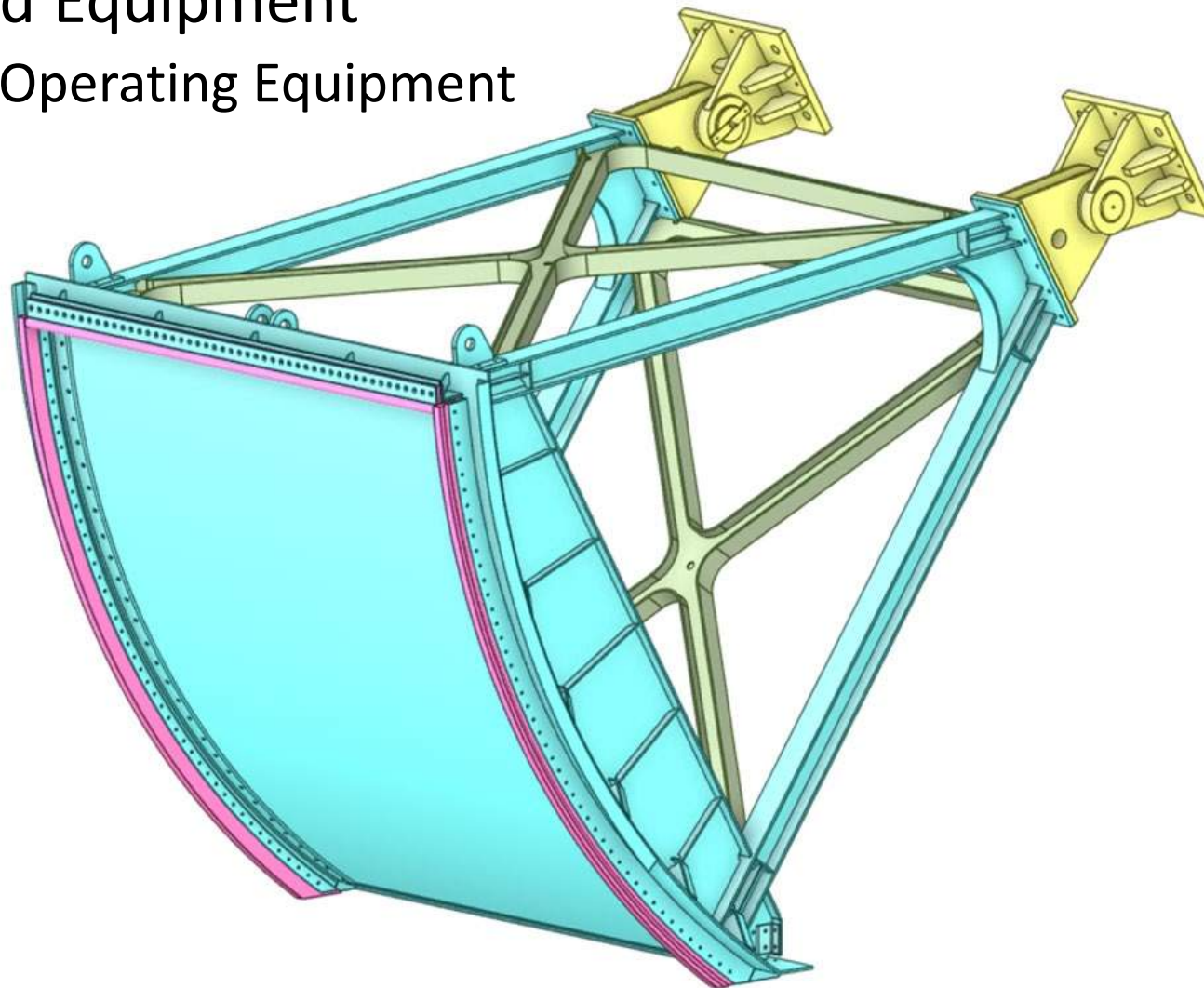
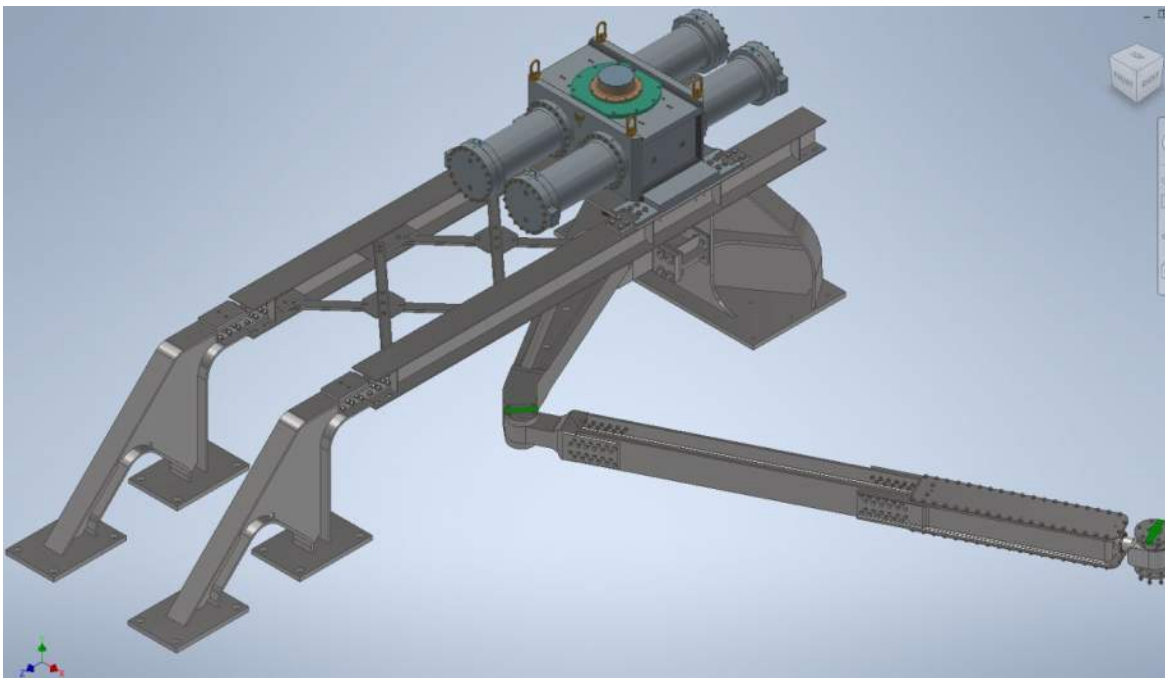
- Installation of Government Furnished Equipment
 - Phase I Hydraulic Steel Structures and Operating Equipment



DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

- Installation of Government Furnished Equipment
 - Phase I Hydraulic Steel Structures and Operating Equipment



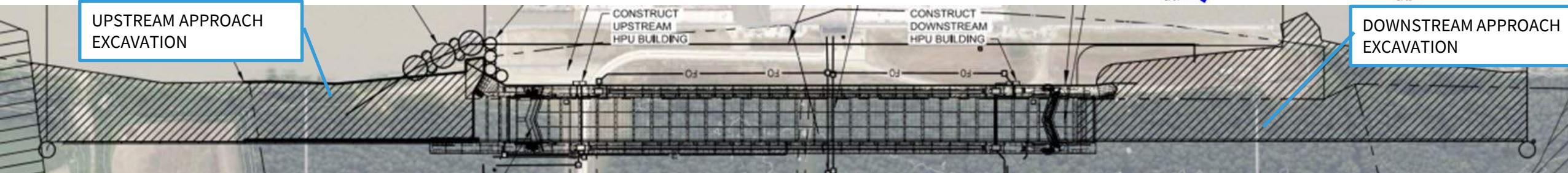
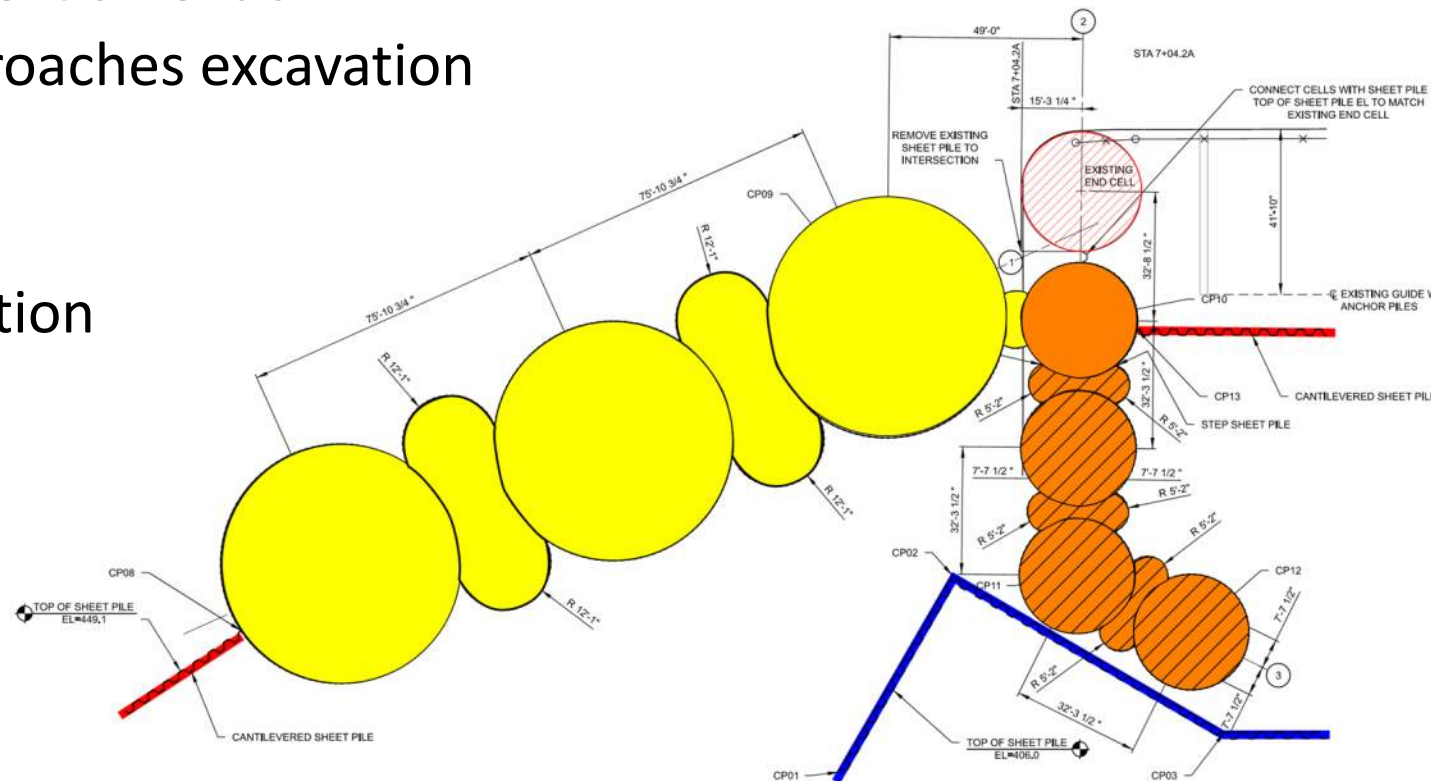
DETAILS OF PHASE IV

LOCK, GUIDE WALLS, AND SUPPORTING FACILITIES

- Controls, Communications, and Monitoring
 - PLC Based Control System
 - Includes Replacement of Control System at Existing Lock
 - Operator Consoles in Each of the Control Station Buildings
 - Ethernet based Communications System with Fiber Optics
 - CCTV System for Operations Monitoring and Security

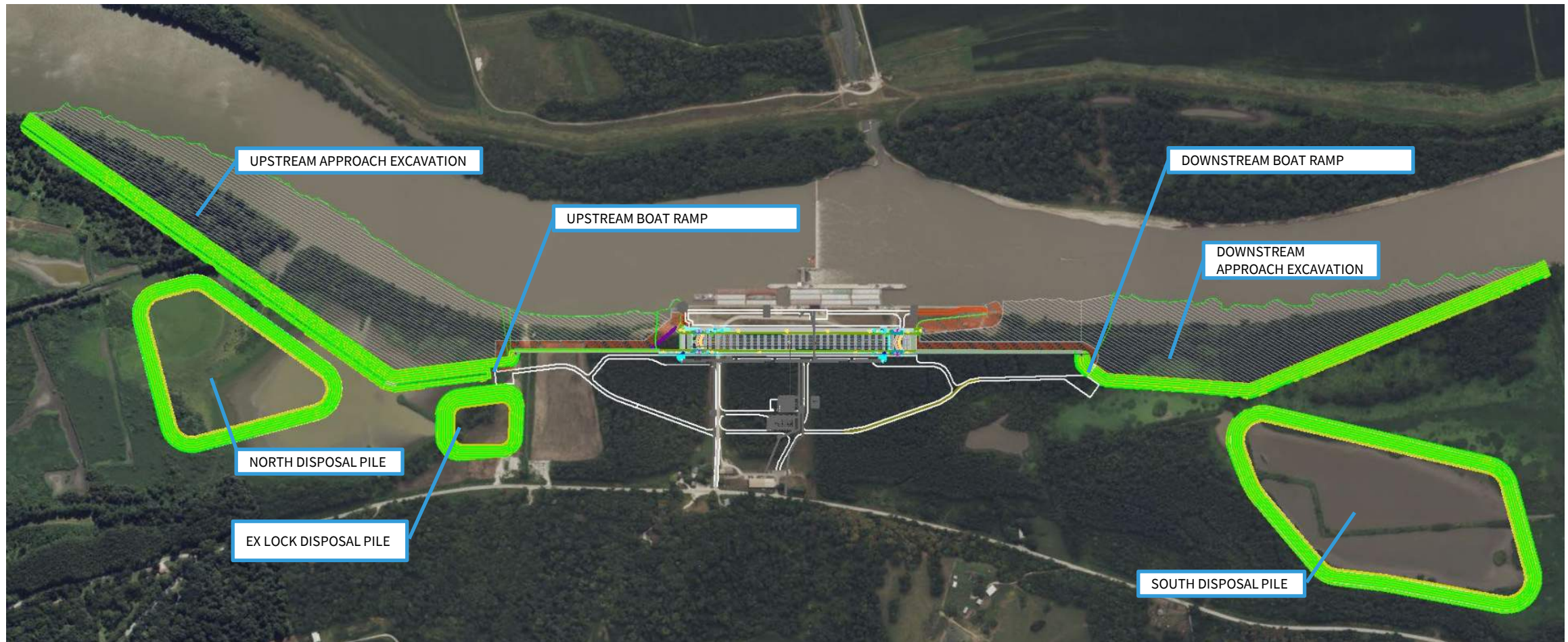
■ Excavation & Dismantling of the Cofferdam

- Upstream and downstream approaches excavation
- Placement of riprap
- Removal of cantilever sheet pile
- Demolition of cells to final elevation
- Removal of cells in the river



PHASE V – APPROACH CHANNELS, SITE WORK AND RESTORATION

- Channel Approach Dredging/Excavation
- Spoil Piles
- Boat Ramps
- Final Site Restoration

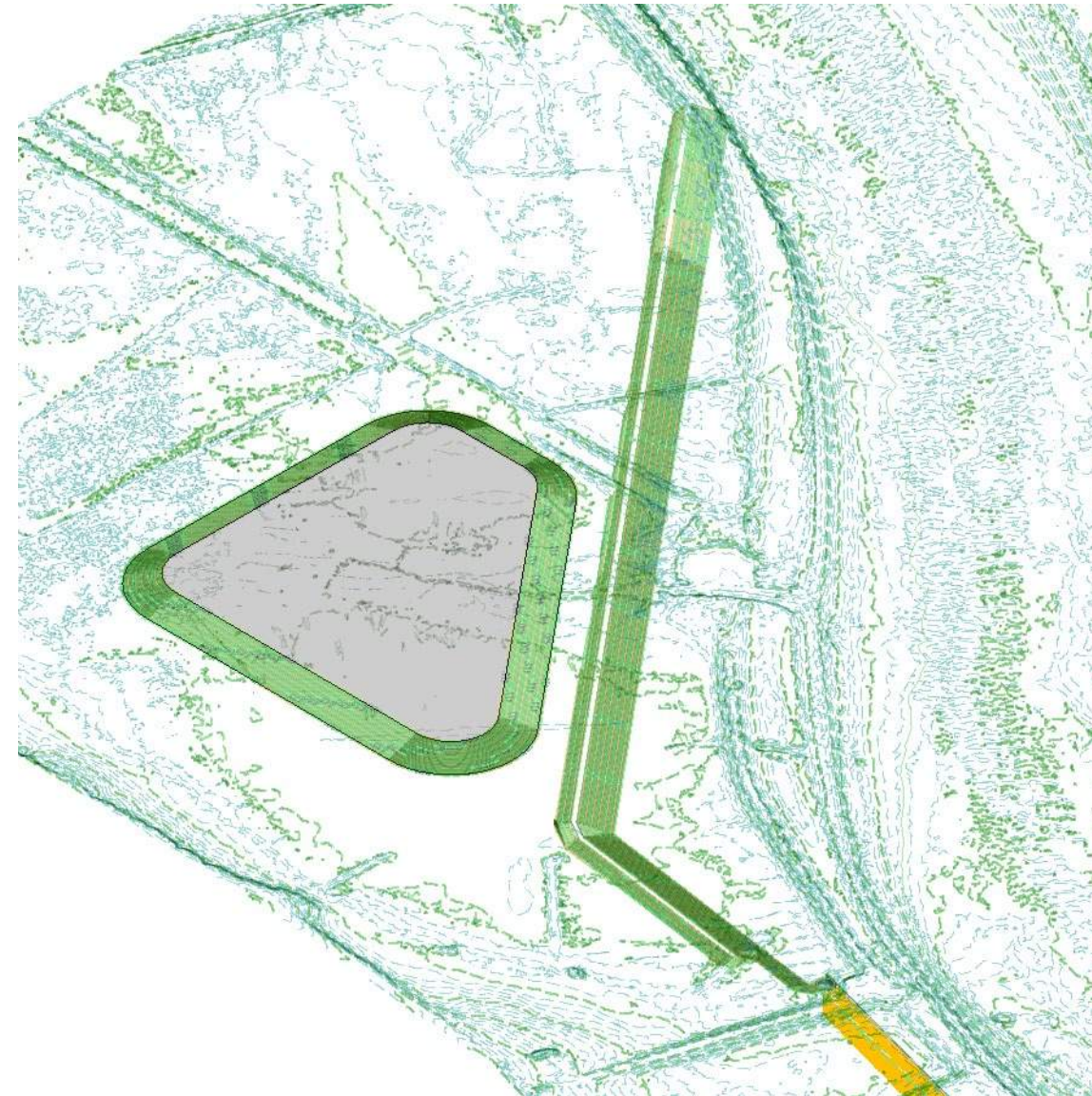


DETAILS OF PHASE V

APPROACH CHANNELS, SITE WORK AND RESTORATION

■ Channel Approach Excavation

- Upstream and Downstream excavation and dredging to construct approach channels to lock constructed in Phase IV
- Current design assumes mechanical excavation methods

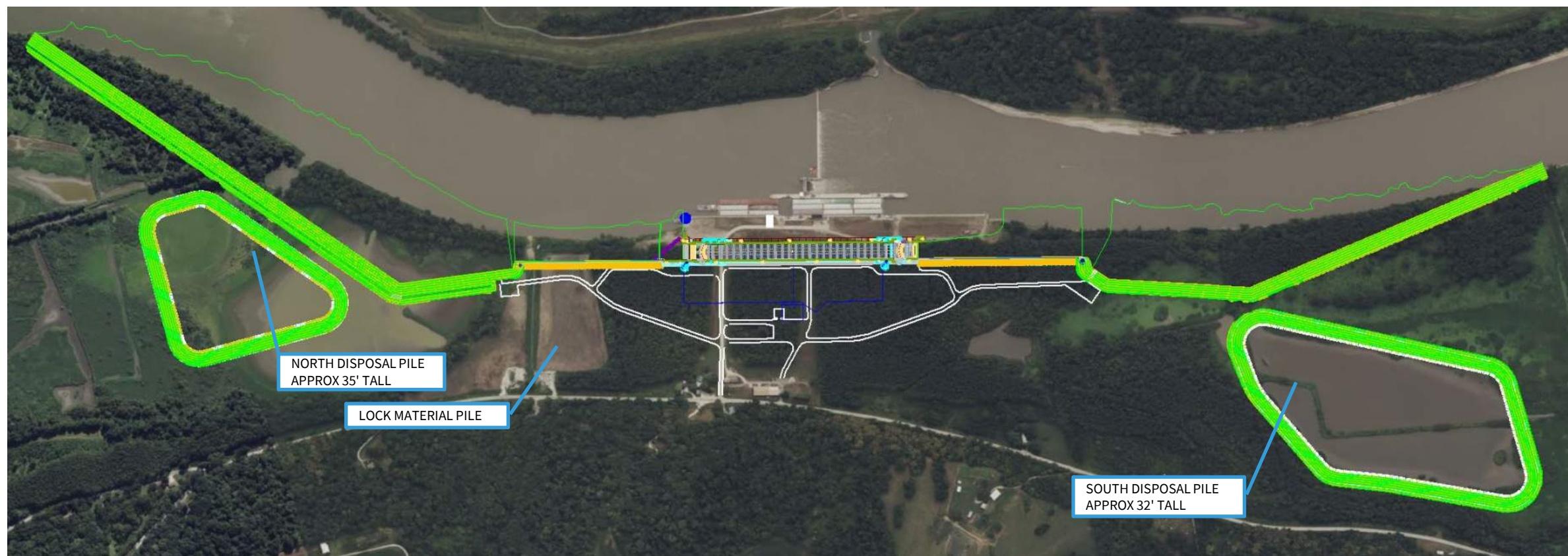


DETAILS OF PHASE V

APPROACH CHANNELS, SITE WORK AND RESTORATION

■ Spoils Piles

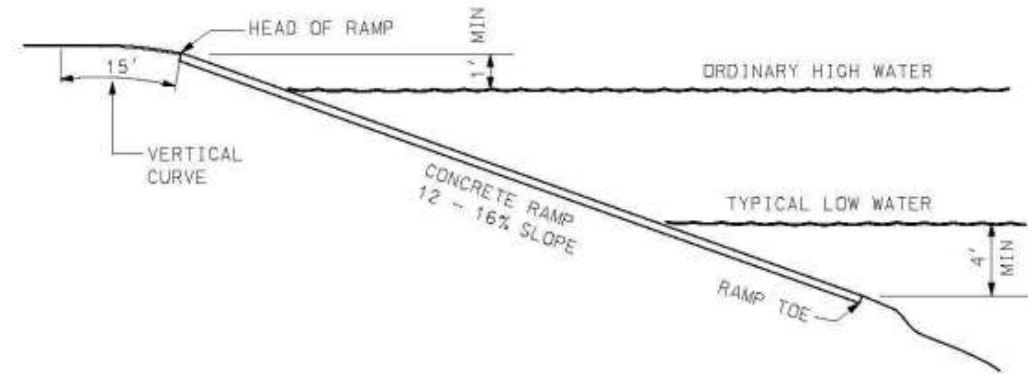
- Excavated materials to be placed at locations on site
- North and south stockpiles
- Approximately 2.35 million cubic yards of material



DETAILS OF PHASE V

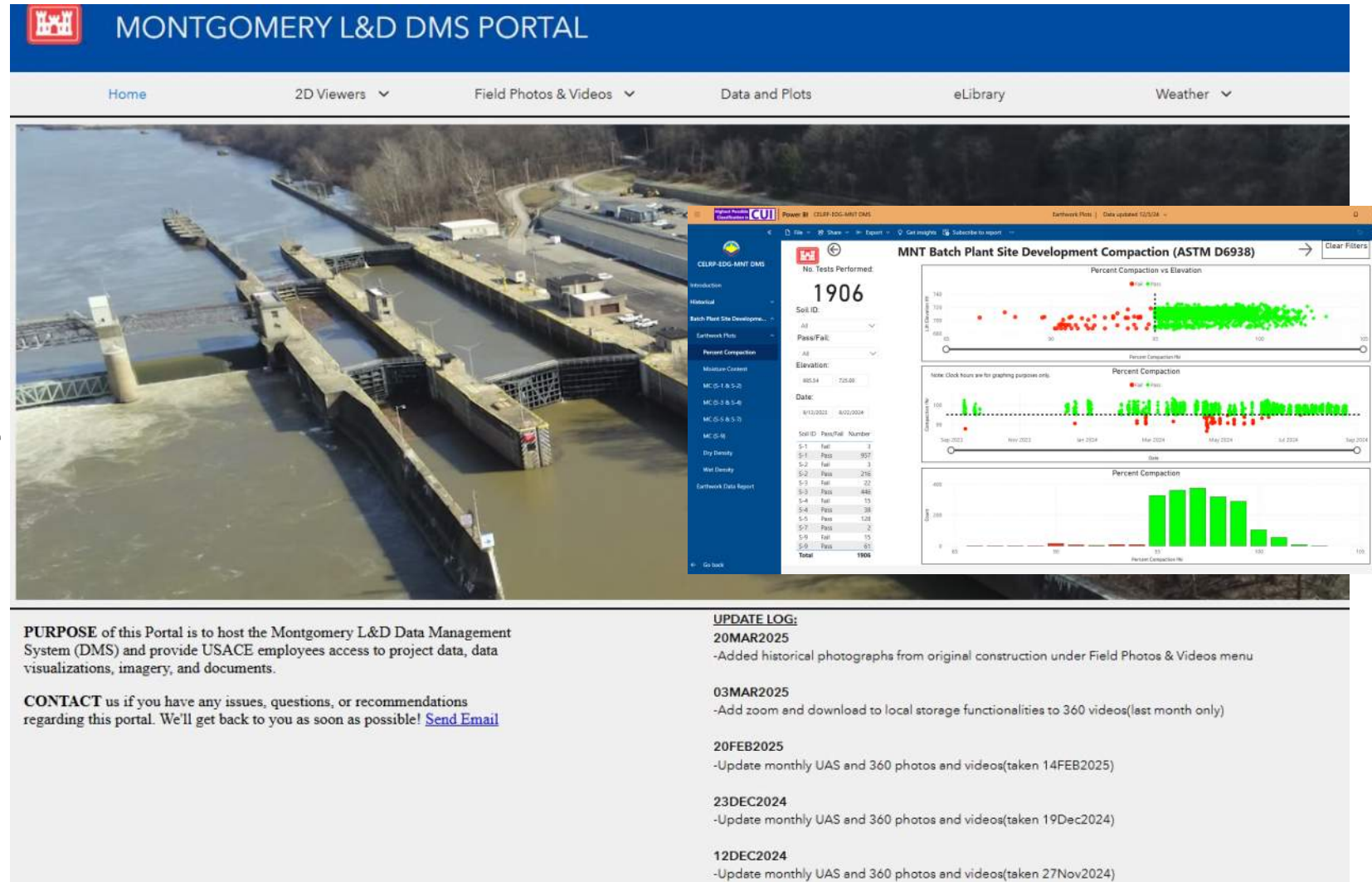
APPROACH CHANNELS, SITE WORK AND RESTORATION

- Work items
 - North Boat Ramp
 - South Boat Ramp
 - Riprap Placement
- Site Restoration
 - Seeding of Spoils Piles
 - Wetland seeding in localized areas
 - Seeding of other disturbed areas



Mega Projects including Montgomery Lock and Soo Lock Require a Data a Management Program

- Specification will state requirements on what to collect, how to store (ASCII, CSV or with MS Excel), how to export thru a secure FTP site, timeframes for presenting the information (from instantaneous to 72 hours)
- Contract will specify a Data Manager and Assistant whose sole responsibility will be to oversee the Contractor's Data Management System stipulated.
- This is in addition to RMS requirements for testing data.
- Allows for storage and access of data in real time during construction.



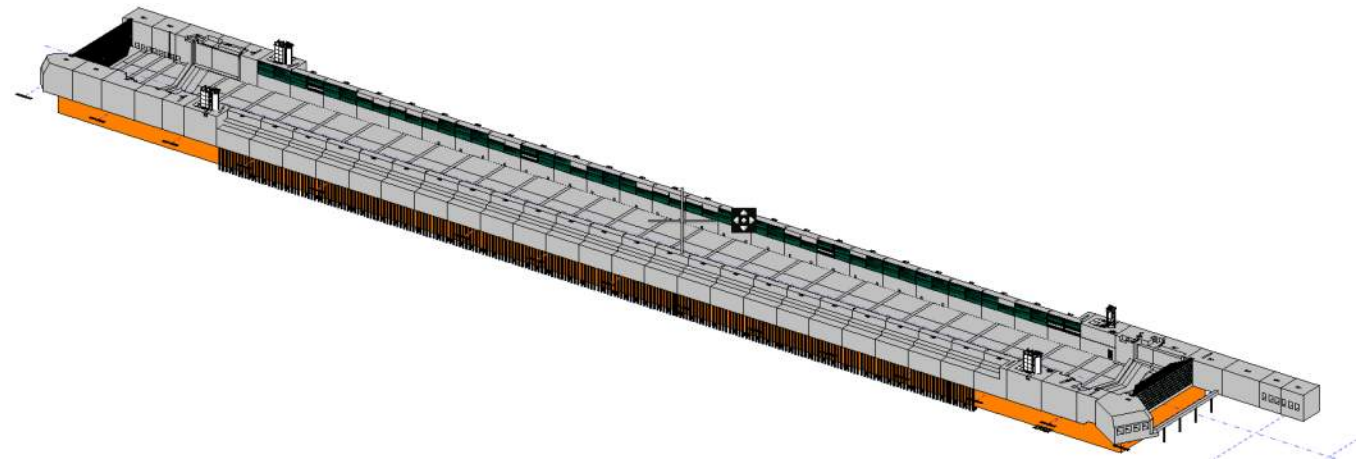
CONCRETE REQUIREMENTS

- Contractor will be required to provide a Thermal Control Plan for Lock Structures
- Concrete placement and curing procedures must meet ACI thermal limits for maximum temperature and maximum temperature differential.

Given that it is likely that the funding will come to this project over several years.

What are some approaches to manage funding shortfalls?

- Multiple Contracts
- Base Contracts – Option periods
- Continuing Contracts/Incremental Funding Clauses



				CY2024	CY2025	CY2026	CY2027	CY2028	CY2029	CY2030	CY2031	CY2032
	Duration (Days)	Cost	Proposed Contract Type	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032
Phase 1 - Procurement of Gates & Operating Machinery	804	\$10M-\$25M	FFP -IFB	■	■	■	■	■	■	■		
Phase 2 - Site Development & Utilities	127	\$10M-\$25M	FFP -IFB	■	■	■	■					
Phase 3 - Lock Excavation, Cofferdam & Foundations	714	\$250M-\$500M	FFP - BV w/Tradeoff	■	■	■	■	■	■	■		
Phase 4 - Guidewalls, Lock & Supporting Facilities	950	\$500M-\$1000M	FFP - BV w/Tradeoff	■	■	■			■	■	■	■
Phase 5 - Approach Channel & Marine Features	468	\$250M-\$500M	FFP - BV w/Tradeoff	■	■	■				■	■	■
Storage Building	365	\$1M-\$5M	FFP -IFB		■	■	■					
Utility Transformers	365	\$500k-\$1M	Utility Contract		■	■	■					

■ Design Fully Funded
 ■ Construction Funded
 ■ Construction Unfunded

FFP-Firm Fixed Price

IFB- Invitation for Bid

BV- Best Value (With Trade Offs or Least Price Technically Acceptable)

MULTIPLE CONTRACTS

				CY2024	CY2025	CY2026	CY2027	CY2028	CY2029	CY2030	CY2031	CY2032
	Duration (Days)	Cost	Proposed Contract Type	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032
Phase 1 - Procurement of Gates & Operating Machinery	804	\$10M-\$25M	FFP -IFB	■	■	■	■	■	■	■		
Phase 2 - Site Development & Utilities	127	\$10M-\$25M	FFP -IFB	■	■	■	■					
Phase 3 - Lock Excavation, Cofferdam & Foundations	714	\$250M-\$500M	FFP - BV w/Tradeoff	■	■	■	■	■	■	■		
Phase 4 - Guidewalls, Lock & Supporting Facilities	950	\$500M-\$1000M	FFP - BV w/Tradeoff	■	■	■			■	■	■	■
Phase 5 - Approach Channel & Marine Features	468	\$250M-\$500M	FFP - BV w/Tradeoff	■	■	■				■	■	■
Storage Building	365	\$1M-\$5M	FFP -IFB		■	■	■					
Utility Transformers	365	\$500k-\$1M	Utility Contract		■	■	■					

■ Design Fully Funded
 ■ Construction Funded
 ■ Construction Unfunded

- Regarding project phasing, what are your thoughts on separating the current phases into individual contracts?
- What are the typical field office overhead rate differences between a \$100 million contract and a \$500 million contract?
- If we divide the project into ten phases, what is the estimated cost impact compared to the current phasing structure?
- What would be the estimated cost savings if we were to combine Phases 3, 4, and 5?

BASE CONTRACTS WITH OPTIONS

Other Project with Base and Options.

Base + Option	Description	Option Period
Contract Base:	Construction of Cofferdam, Land wall and River wall.	N/A
Option 1:	Guard walls	1095 Days (3 Years)
Option 2:	Sills and Floors	1460 Days (4 Years)
Option 3:	Hydraulic Steel Structures: Gates, Valves Bulkheads etc,	1460 Days (4 Years)
Option 4:	Buildings, Lock electrical and Mechanical Equipment (Plumbing Systems)	1825 Days (5 Years)
Option 5:	Removal of the Cofferdam	2555 Days (7 Years)
Option 6:	Site and Groundwork, road paving, site Restoration.	365 Days

- What premium is typically applied to optioned items in a contract?
- What are the advantages and disadvantages of including option periods in contracts, from both the client's and contractor's perspectives?
- What is the typical duration of option periods in your experience? At what point in an option period's length do contractors typically begin to factor significant risk premiums into their pricing?
- Are there alternative approaches or best practices for improving option pricing mechanisms, especially for multi-year options, that could mitigate risk and incentivize both parties?

CONTINUOUSLY & FULLY FUNDED CONTRACTS

- Possible Funding outline shown below:
 - What are the advantages of incremental funding
 - What are the concerns:

				CY2024	CY2025	CY2026	CY2027	CY2028	CY2029	CY2030	CY2031	CY2032
	Duration (Days)	Cost	Proposed Contract Type	FY2024	FY2025	FY2026	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032
Phase 1 - Procurement of Gates & Operating Machinery	804	\$10M-\$25M	FFP -IFB									
Phase 2 - Site Development & Utilities	127	\$10M-\$25M	FFP -IFB									
Phase 3 - Lock Excavation, Cofferdam & Foundations	714	\$250M-\$500M	FFP - BV w/Tradeoff									
Phase 4 - Guidewalls, Lock & Supporting Facilities	950	\$500M-\$1000M	FFP - BV w/Tradeoff									
Phase 5 - Approach Channel & Marine Features	468	\$250M-\$500M	FFP - BV w/Tradeoff									
Storage Building	365	\$1M-\$5M	FFP -IFB									
Utility Transformers	365	\$500k-\$1M	Utility Contract									

■ Design Fully Funded
 ■ Construction Funded
 ■ Construction Unfunded

Expenditures (Millions)

	Start	Finish	2024	2025	2026	2027	2028	2029	2030	2031	2032	Totals
Phase 1	01-Oct-24	30-Oct-27	\$2	\$8	\$8	\$6						\$24
Phase 2	31-Oct-25	27-Apr-26		\$8	\$15							\$24
Phase 3	28-May-26	20-Feb-29			\$48	\$80	\$81	\$11				\$220
Phase 4	21-Feb-29	12-Oct-32						\$184	\$214	\$214	\$168	\$780
Phase 5	05-Nov-29	20-Aug-31						\$22	\$139	\$89		\$250
Totals			\$2	\$16	\$71	\$87	\$81	\$217	\$353	\$303	\$168	\$1,297
Fully Funded			\$24	\$24	\$220			\$1,030				

- Does the construction sequencing/approach seem reasonable? Any suggestion to make the approach more efficient?
- Contract acquisition types?
 - Best Value with Tradeoffs (BV)
 - Lowest Price Technically Acceptable (LPTA)
 - Invitation for Bid (IFB)
- How long would you require (30/45/60 days) to prepare a responsive proposal to a construction contract solicitation?
- What do contractors see as the largest risks with this project(s)?
 - What some mitigation approaches

Our goal is to prepare contract plans that are buildable and biddable.

Contractors who have bid on plans, have worked on USACE projects, or are considering bidding on the future LaGrange Lock plans are being asked to provide feedback that will aid in this goal.

Please consider answering the following questions to provide information that will help us and may help you.

- The second transformer installed under the contract work will be used to provide the power needed by the contractor for all construction activities.
 - The bill for the power will be paid by the contractor.
 - What is the maximum usage anticipated by the contractor that the designers should use to size this transformer?
 - Some of the major construction loads include: construction trailers, well water pumps, dewatering pumps, concrete batch plant, guide wall pumps, and central air compressors.

- Does the construction sequencing/approach seem reasonable? Any suggestion to make the approach more efficient?
- Concrete materials availability?
- Driving flat sheet piles ~80ft?
- Flood risk?
 - Cofferdam height vs flood recovery effort
- What do contractors see as the largest risks with this project(s)?
- How often should we have industry days?
 - Should we have online and face to face industry days?